INCH-POUND

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PERFORMANCE SPECIFICATION

BATTERIES, RECHARGEABLE, SEALED

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1 SCOPE

1.1 Scope. This specification covers sealed rechargeable batteries designed to power portable communications electronics devices used by the US military. Detailed electrical and physical requirements for individual batteries are specified on specification sheets.

1.2 Classification. Batteries described by this specification have the following types:

Type I – Batteries with a nickel cadmium (Ni-Cd) electrochemical system

Type II – Batteries with a nickel metal hydride (Ni-MH) electrochemical system

Type III – Batteries with a lithium ion (Li-ion) electrochemical system (see 6.4.8)

Part or identifying number (PIN). PINs to be used for batteries described by this specification are created as described in TABLE I below. Additional differentiation may be required (see 3.1).

TABLE I. *PIN description*

M	32383	/1	-3[1]
M prefix	Specification number	Specification sheet number	Type

NOTES

[1] Roman numeral type numbers are converted to Arabic digits in the PINs to provide a consistent number of characters in the PINs assigned by this specification.

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Comments, suggestions, or questions on this document should be emailed to usarmy.apg.ccdc-c5isr.mbx.standardization-crx@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DoD SPECIFICATIONS

MIL-DTL-83133	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8), NATO F-35, and JP-8 + 100

(See ASSIST database for list of specification sheets.)

DoD STANDARDS

MIL-STD-130	DoD Standard Practice: Identification Marking Of U.S. Military Property
MIL-STD-882	DoD Standard Practice System Safety
MIL-STD-810	DoD Test Method Standard: Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil/)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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NAVSEA S9310-AQ-SAF-010 Technical Manual for Batteries, Navy Lithium Safety Program Responsibilities and Procedures

(Copies of this document may be obtained from the Naval Ordnance Safety and Security Activity (NOSSA), Farragut Hall Building D-323, 23 Strauss Avenue, Indian Head, MD 20640-5555.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

IEC 61000-4-2 Electromagnetic compatibility (EMC) – Part 4-2: Testing and measurement techniques - Electrostatic discharge immunity test

(Copies of this specification are available online from International Electrotechnical Commission https://webstore.iec.ch)

ASTM G115-10 Standard Guide for Measuring and Reporting Friction Coefficients (Copies of this specification are available online from ASTM International https://astm.org)

ASTM-D-1141 Standard Practice for the Preparation of Substitute Ocean Water

(Copies of these documents are available online from https://astm.org)

System Management Bus Specification Version 1.1

Smart Battery Data Specification Version 1.1

System Management Bus, SMBus, Smart Battery, Smart Battery System, SBS, SMBus and SBS logos are trademarks of the System Management Interface Forum (SMIF), Inc.

(Copies of these documents may be obtained from the SMIF, Inc, online at http://powersig.org)

Universal Serial Bus Power Delivery (USB PD) Specification 3.1

The USB Type-C Connectors and Cable Assemblies Compliance Document v 2.0

USB Power Delivery Compliance Test Specification Version 1.0 Revision 1.2, September 2020

(Copies of these documents may be obtained from https://usb.org)

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SAE-AMS-STD-595 Colors Used in Government Procurement

(Copies may be obtained from https://sae.org/)

UL 94 Standard for Safety, Test for Flammability of Plastic Materials for Parts in Devices and Appliances

UL 1642 Standard for Safety, Lithium Batteries

(Applications for copies should be addressed at https://www.ul.com/).

UN Manual of Tests and Criteria

(Copies of this document may be obtained at https://un.org)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern (see 6.5).

3.2 First Article requirements. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

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3.2.1 Certifications. Certification of conformity shall be provided for the following requirements. The Government reserves the right to verify any or all certifications.

TABLE II. *Certifications of conformity*

REQUIREMENT	PARAGRAPH
Metals	3.3.1
Dissimilar metals	3.3.2
Resistance of elastomeric materials	3.3.5
Electrical connection wires and tabs	3.3.6
Electrical contacts	3.4.4.1
Connector	3.4.4.1.1
Battery case[1]	3.4.5
Display window[2]	3.4.5.1
Dust cap or cover	3.4.7.1
Display, smart[3]	3.5.8.2
SMBus[4]	3.5.9.1
Battery wake up[5]	3.6.8.2
Battery Safety	3.7.2
High temperature temporary cutoff devices[6]	3.7.2.4
Inter-cell protective devices	3.7.2.6
Redundant safety protection	3.7.2.7
Transportation requirements	3.9
Connector pull force	3.4.4.1.1.1

NOTES
[1] UL Standard 94 – flammability
[2] Transparency during battery life
[3] Requirements b and g
[4] Charge enable
[5] When battery sleep mode is utilized
[6] Number of devices

3.3 Materials.

3.3.1 Metals. All metals which do not enter into the basic electrochemical reaction of the cell shall resist, or be treated to resist, corrosion. Certification is required.

3.3.2 Dissimilar metals. When dissimilar metals which would adversely affect battery performance are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. Certification is required.

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3.3.3 Corrosion resistance. External parts of the battery shall show no evidence of cracking, pitting, chipping, scaling, corrosion or other deleterious effects after being tested as specified herein.

3.3.4 Insulating compounds, flow, and shrinking. Insulating, impregnating, potting and sealing compounds used in manufacturing shall be capable of performing their intended purpose under the use conditions described by this specification. Compounds shall not inhibit the operation of any safety features. When tested as specified in 4.7.1.5, the insulating, impregnating, potting and sealing compounds shall not flow at high temperature, and shall not crack or draw away from the sides of a container at low temperature. Any compound used shall be non-flammable and non-toxic (see 6.4.9).

3.3.5 Resistance of elastomeric materials. All elastomeric materials used in the battery shall show no cracks, blisters or other deterioration, nor cause degradation of battery performance after being tested as specified herein. Any elastomeric material used shall be non-flammable and non-toxic (see 6.4.9). Certification is required.

3.3.6 Electrical connection wires and tabs. All points inside a battery that have positive and negative polarity in close proximity (i.e. posing a risk of short circuit throughout the life of the battery) shall have not less than one layer of insulation between the positive and negative. Insulating material, or layers of multiple insulating materials between positive and negative points, shall have the following characteristics as a minimum:

a. Softening temperature (see 6.4.11): Not less than 302°F (150°C).
b. Lengthwise shrinkage: Not greater than 3% after application
c. Thickness: Not less than 0.005 inch
d. Material shall not shrink, soften, or crack during any of the tests of this specification.
e. The material shall be non-flammable, non-toxic (see 6.4.9), and impervious to the electrolyte of the battery.
f. Certification is required.

3.3.7 Additional materials requirements. When specified (see 3.1), batteries shall be subjected to additional materials requirements.

3.4 Physical characteristics.

3.4.1 Visual and mechanical. Each battery shall be free of the visual mechanical defects listed in TABLE XIV.

3.4.2 Cells. The battery shall be made up of the necessary number of individual series and/or parallel connected cells, which have demonstrated their capability to meet the specified electrical and safety requirements.

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3.4.2.1 Protective coverings. Metallic cells used to build batteries shall have an individual protective sleeving or coating. Material selected shall meet the requirements of 3.3.4; and 3.3.5 if applicable.

3.4.3 Dimensions and weight. The dimensions and weights of each battery shall be as shown on the applicable specification sheets.

3.4.4 Battery connector and other electrical interfaces.

3.4.4.1 Electrical contacts. All electrical contacts of each battery shall be resistant to corrosion and shall be capable of withstanding the required number of insertions or connections specified herein (see 3.4.4.1.1.2 and 3.4.4.1.2). Each contact shall consist of the base metal and finish specified in the applicable specification sheet. Compliance to base metal and finish requirements shall be certified.

3.4.4.1.1 Connector. Unless otherwise specified (see 3.1), the battery shall contain a connector as shown on the applicable specification sheet.

3.4.4.1.1.1 Connector pull force. When specified (see 3.1), the battery connector shall have the specified disconnect pull force. Certification is required.

3.4.4.1.1.2 Connector insertions. When specified (see 3.1), the battery connector shall maintain a fully functional interface after 500 insertions and extractions. The final insertion and extraction forces shall be within 15% of the initial insertion and extraction forces. When tested as specified in 4.7.1.3, battery connector shall not tear, rip, be displaced, or separate from the battery case.

3.4.4.1.2 Flat terminal. The battery shall contain flat terminals as specified in the applicable specification sheet. Battery terminals, including data and charge terminals, shall be of sufficient strength to withstand 250 connections without damage that would interfere with connections and shall maintain a fully functional interface. When tested as specified in 4.7.1.4, battery terminals shall not tear, rip, be displaced, or separate from the battery case.

3.4.4.1.3 Multilayer finish. When specified (see 3.1), electrical interfaces shall be finished according to the following:

a. Electrolytic copper flash, not less than 99.5% pure, thickness 10-20 micro-inches.
b. Electroless nickel plate; not less than 800 Knoop hardness; not less than 200 micro-inches thickness.
c. Gold plate, not less than 99% pure; 130-200 Knoop hardness; not less than 30 micro-inches thickness.

3.4.4.1.4 Gold plating. When specified (see 3.1), electrical interfaces shall be finished according to the following: Gold plate, not less than 99% pure, 91-129 Knoop hardness, 25 micro-inches thick or better.

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3.4.4.1.5 USB power port connector(s). When specified (see 3.1), the battery shall include the specified number and type(s) of USB connector(s). Location(s) shall be as specified.

3.4.4.1.5.1 USB type A connector. The USB connector shall have the following characteristics (see 4.7.2.17):

a. One USB Dedicated Charging Port (DCP) capable of supplying 1.5 A of current.
b. A tethered protective cover.
c. A charger adapter emulator circuit to accommodate proprietary fast charging protocols and be interoperable with Apple devices, Samsung devices, or devices containing Qualcomm Quick Charge.
d. The USB power port connector(s) shall be located as specified (see 3.1).

3.4.4.1.5.2 USB type C connector. The USB connector shall have the following characteristics (see 4.7.2.17):

a. Not less than 3 Amps at specified (see 3.1) VDC female port (receptacle) shall be available.
b. A tethered protective cover.
c. The USB power port connector(s) shall be located as specified (see 3.1).
d. The USB power port shall support dual-role power capabilities as specified in the USB PD specification (See 3.5.9.2).
e. The USB power port shall use a certified USB Type-C (USB 2.0 or higher) receptacle per the USB Type-C Connectors and Cable Assemblies Compliance Document v 2.0 per power delivery 3.0.

3.4.5 Battery case. The battery case shall be capable of maintaining the specified dimensions during the life of the battery. The surface of the case shall have a smooth finish free from pitting, blow-holes, rough spots, or other deformations. The battery case edges, seams, and corners shall be smooth. The case shall be fabricated of material having sufficient strength to withstand the environmental and electrical tests specified herein. Material shall be nonflammable. When plastics are used for the battery case, the material shall be classified in accordance with UL Standard 94, Test for Flammability of Plastic Materials for Parts in Devices and Appliances, except as otherwise noted herein. Acceptable ratings include: HB; V-0; V-1; V-2; 5VA; 5VB; VTM-0; VTM-1; and VTM-2. If the manufacturer elects to use material classified in accordance with UL 94, certification is required. If the manufacturer elects to use a plastic material that doesn't have UL-94 classification, then the test of 4.7.1.7 shall be performed. When tested as specified in 4.7.1.7, test specimens shall self-extinguish within 5 seconds after removal from flame.

3.4.5.1 Display window. The window shall remain intact, in place, and transparent during the life of the battery. The window shall have a smooth finish free from pitting, blow-holes, rough spots, or other deformations and shall be fabricated of material having sufficient strength to withstand the environmental and electrical tests specified herein. Certification for transparency during the life of the battery is required.

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3.4.5.2 Battery case expansion. The battery shall be designed to prevent dimensional expansion that will prevent its removal from equipment.

3.4.5.3 Battery case vent. A pressure release device shall be incorporated into the battery case as specified (see 3.1). This device shall be designed to release below the bursting pressure of the container and above the normal operating pressure of the battery. The design of the battery shall be such that the paths from the individual cell units to the battery vent are sufficient to permit the removal of gases. The device shall not permit the entry of liquids into the battery case. When tested as specified in 4.7.1.6, each battery case shall not exhibit any signs of visible damage other than the case vent opening, and each battery shall pass through the test fixture with a force not greater than 12 pounds.

3.4.5.4 Color. The color of the battery case shall be as specified below, in accordance with SAE-AMS-STD-595:

a. Type I (Ni-Cd) and Type II (Ni-MH) batteries shall be lusterless green, chip number 34088.
b. Type III (Li-ion) batteries shall be lusterless “Tan 686A”, chip number 33446.

3.4.5.5 Writable label. A writable/erasable label shall be attached to each battery. This rectangular label shall state "User Info" and contain an area that allows writing and erasing multiple times. The label shall be not less than 1 inch X 13/16 inch.

3.4.5.6 Surface friction. Batteries shall not interfere with the operation of the system. When specified (see 3.1), batteries shall have the specified surface friction when tested in accordance with 4.7.1.8.

3.4.6 Marking. Each battery shall be provided with a permanent, legible hot-stamped or printed marking as specified herein. Direct printing on the battery is the preferred method. For rectangular batteries, the marking may be placed on more than one surface. Lettering shall be without serif (sans-serif) fonts. Unless otherwise specified (see 3.1), the font for all identification and instruction markings shall be not less than 0.1 inch (2.5 mm).

3.4.6.1 Identification marking. The identification marking shall contain the following information. The information in parentheses shall be filled in by the manufacturer.

a. BATTERY, RECHARGEABLE, (Type Designation, PIN)
b. (Chemistry)
c. (Battery Voltage) V, (Full Capacity Rating) Ah / (Full Energy Rating) Wh
d. (NSN)
e. (Contract Number)
f. (MFG DATE) (Serial Number)
g. (MFG Warranty Details)
h. (Manufacturer)
i. (Manufacturer’s location)

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3.4.6.1.1 Item unique identification (IUID). When specified (see 3.1), batteries shall contain IUID marking in accordance with MIL-STD-130.

3.4.6.2 Warning marking. Unless otherwise specified (see 3.1), the following warning marking shall appear on each battery in capital letters. The font shall not be less than 0.16 inch (4 mm).

WARNING/STORAGE
DO NOT STORE ABOVE 122°F (50°C),
CRUSH, MUTILATE, REVERSE POLARITY,
DISASSEMBLE, OR DISPOSE OF IN FIRE.

3.4.6.3 Manufacture date. The code shown shall indicate the month and year of manufacture of the battery by means of a four-digit number in which the first two digits shall indicate the number of the month and the last two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "0". A forward slash (“/”) shall separate the first two digits from the last two (e.g. 01/13 indicates January 2013). When a battery is completed during the last three working days of a month or the first three working days of the subsequent month, the manufacturer is permitted to use either month as the coded month of manufacture.

3.4.6.4 Terminal marking. Positive terminals shall be legibly indicated by "+", and negative terminals shall be identified by "-". The polarity markings shall be placed as close as possible to the applicable side of the receptacle where required. A schematic drawing shall be placed as close as possible to the receptacle or terminals as shown on the applicable specification sheet (see 3.1).

3.4.6.5 Instructions/notes. Each battery shall be marked with complete instructions for operation and charging of the battery, which shall include the following:

a. (Preferred charging method)
b. Specific instructions can be found in the applicable specification sheet (see 3.1).
c. (Alternate charging method)
d. Specific instructions can be found in the applicable specification sheet (see 3.1).
e. State of charge indicator (SOCI) matrix (see 3.5.8)
f. Self-discharge rate
g. Maintenance (e.g. discharging and charging periodically) where applicable
h. Other notes when specified (see 3.1)

3.4.6.6 Simplified battery label. The simplified battery label is designed to provide the end user with an easily understandable label containing the most pertinent information. The label size and location shall be as specified (see 3.1). Numeric values shall be for a complete battery, in new condition, at room temperature. The entire label shall be framed by 2 concentric arrows, indicating a rechargeable battery. The statement “For more information see

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https://battery.army.mil" shall be printed below the label. See FIGURE 1 for a sample representation of the simplified battery label.
The simplified battery label shall contain the following:

a. Type designation
b. Total energy in watt hours (rounded to nearest whole number), presented inside a battery icon
c. The total weight in pounds (rounded to the nearest 0.1 pounds), presented inside a barbell icon
d. Energy density shall be calculated in watt hours per kilogram. The value shall be rounded to the nearest 50 Wh/kg, and one star icon shall be presented for every 50 Wh/kg. The actual numeric value (rounded to nearest whole number) shall be placed below the stars.
e. Battery chemistry (see 3.1)

For more information see https://battery.army.mil.

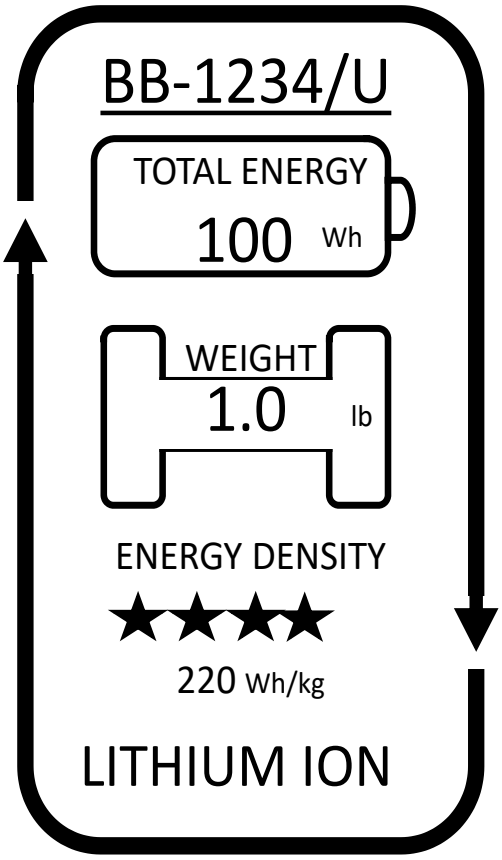


FIGURE 1. Sample simplified battery label
Note: Battery nomenclature and values are fictitious

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3.4.6.7 Color. The color of the marking shall be as specified below:

a. Type I (Ni-Cd) batteries – black
b. Type II (Ni-MH) batteries – white
c. Types III (Li-ion) batteries – black

3.4.6.8 Permanency and durability. Marking shall be capable of withstanding all tests of 4.7, except projectile (see 4.7.4.11), without evidence of any of the following:

Blistering	Delamination	Separation	Discoloration
Chipping	Dissolving	Softening	Illegibility
Corrosion	Loosening	Splitting	Flaking
Cracking	Peeling	Warping	Fading

3.4.6.9 Additional markings. When specified (see 3.1), batteries shall be subjected to additional markings.

3.4.7 Battery condition for shipping. All batteries shall be supplied with instructions cards (see 3.1) and a dust cap or cover over the connector.

Types I and II Batteries: Each battery shall be furnished fully charged as indicated by five state of charge segments where applicable.

Type III Batteries: Each battery shall be 30% state of charge (SOC) +0/-10% and indicate two state of charge segments (see TABLE III).

3.4.7.1 Dust cap or cover. Each battery shall be furnished with a dust cap for shipping/transportation. The dust cap or cover shall not fall off during normal handling and shall be removable by hand at temperatures from -4°F to 131°F (-20°C to 55°C). The material used for the dust cap shall be non-toxic and non-flammable and shall withstand temperatures from -40°F to 195°F (-40°C to 90°C) without shrinkage or cracking. The dust cap or cover shall not leave any residue on the battery contacts nor have any adverse effect on the connection interface. Dust cap or cover compliance to these requirements shall be certified (3.2.1).

3.4.8 Additional physical characteristics requirements. When specified (see 3.1), batteries shall be subjected to additional physical characteristics requirements.

3.5 Electrical requirements.

3.5.1 Cell balance. Cells used to build each battery shall be matched to the limits specified below in order to avoid shortened cycle life. When tested as specified in 4.7.2.1, the capacity of each cell tested shall not vary from the average capacity obtained by more than the following specified limits:

a. Type I (Ni-Cd) and Type II (Ni-MH) cells: ± 5%
b. Type III (Li-ion) cell: ± 2.5%

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3.5.2 Battery voltage. The open circuit voltage (OCV) shall be within the voltage range specified (see 3.1).

3.5.3 Capacity. The discharge in amperes hours or watt hours as required (see 3.1) for each discharge shall be not less than the value listed in the applicable specification sheet. The minimum cutoff voltage shall be the value listed in the applicable specification sheet. The batteries shall meet the specified minimum requirements. The surface temperature of the battery shall not exceed 185°F (85°C).

3.5.4 Cycle life. When tested as specified in 4.7.2.4 batteries shall be capable of retaining the specified capacity during and after the specified number of cycles (see 3.1).

3.5.5 Inrush current. When specified (see 3.1), batteries shall be capable of providing inrush currents required by the system. When tested as specified in 4.7.2.12, battery voltages shall not fall below the specified value. After completion of the test, batteries shall meet the capacity requirements of 3.5.3 and the criteria of TABLE XIV.

3.5.6 Battery charging.

3.5.6.1 Charging interface. The battery shall have the specified charging interface (see 3.1).

3.5.6.2 Charge enable. When specified (see 3.1), batteries shall be equipped with a charge enable circuit that allows charging (see 3.1) on approved chargers (see 6.6) and disallows charging when connected to a charger that doesn't recognize the charge enable circuit protocol. Whenever a charger doesn't recognize the charge enable protocol, the battery shall terminate any charge with current that exceeds the maximum specified (see 3.1). Unless otherwise specified, the charge enable pin shall have an equivalent input circuit of a resistor of specified value in series with a diode of specified VF (see 3.1). The circuit shall require activation current not less than the value specified (see 3.1). When tested as specified in 4.7.2.6, (1) the charge current shall not exceed the value specified without a charge enable circuit protocol being applied, and (2) battery current shall meet the preferred charging method while the charge enable is connected to a charge circuit equipped to allow the preferred charging method if the charge enable terminal or socket meets specified requirements (see 3.1).

3.5.7 Power consumption. The power consumption for all electronics within the battery shall be less than 350 micro-ampere average, per battery or independent section where applicable, at 68 ± 5°F (20 ± 2.8°C) (See 4.7.2.13).

3.5.8 Display. Batteries shall display their status to the user. The value displayed shall compensate for use temperature and discharge rate of the battery.

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3.5.8.1 Display, non-smart. When specified (see 3.1), the battery shall have a non-smart display. The display size and location shall be as specified (see 3.1). The display shall have five distinct segments showing the ranges of state of charge of the battery. Dual section batteries require a separate display for each section. As the battery ages and experiences capacity fade from use, the display shall compensate by showing the fully charged state as a percentage of the original fresh capacity (commonly referred to as “absolute state of charge”). For example, if the original battery capacity is 10 Ah, and as a result of aging it is now a maximum of 8 Ah, a fully charged battery will show four bars and not five. The state of charge ranges displayed shall be as follows (see 4.7.2.14):

TABLE III. *State of charge ranges*

SEGMENTS	STATE OF CHARGE
0	0% (Fully Discharged)
1	1 to 20%
2	21 to 40%
3	41 to 60%
4	61 to 80%
5	81 to 100%

3.5.8.1.1 Display, non-smart, always on. When specified (see 3.1), the battery shall have an always-on SOCI. The segments shall be black in color on a white or pale gray background. The display shall be on continuously with no user action required except that it may go blank temporarily as it refreshes the display.

3.5.8.1.2 Display, non-smart, user activated. When specified (see 3.1), the battery shall have a user activated SOCI. The display shall be activated as specified. The display shall remain active for 3 seconds ± 1 second. The segments shall be illuminated green and shall be compatible with Night Vision Imaging Systems. The output of the display, measured in a direction normal to the display shall be 1.0 to 10.0 millicandelas. Certification of light intensity is required.

3.5.8.2 Display, smart. When specified (see 3.1), the battery shall have a user activated SOCI. The display size and location shall be as specified (see 3.1). See 4.7.2.14.2.

The display shall have the following characteristics:

a. The display shall be a 6 character display which shall indicate the following fields:

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TABLE IV. *Display content*

Fields	Character 1	Character 2	Character 3	Character 4	Character 5	Character 6
SOC			Number	Number	Number	%
State of Health (SOH)			Number	Number	Number	H
Run-Time-To-Empty (RTTE)	Number (Hour)	Number (Hour)	:	Number (Minute)	Number (Minute)	E
Charge-Time-To-Full (CTTF)	Number (Hour)	Number (Hour)	:	Number (Minute)	Number (Minute)	F
Battery Cycle Count (BCC)	Number	Number	Number	Number	C	Y
Unusable condition	0	0	0			

b. Display segments shall be white on a gray background.
c. The output of the display shall be legible from a viewing angle of not less than 45 degrees in any direction. The viewing angle shall be certified.
d. There shall be an actuator button to the right side of the display that is recessed to protect from inadvertent operation. The operations of the actuator button shall have four possible inputs as follows:
    i. Single quick depression. (≤ 6 seconds)
    ii. Multiple quick depressions. (≤ 6 seconds each)
    iii. Long depression. (> 6 seconds)

    The display shall remain illuminated from 3 to 5 seconds following a depression. The actuator shall reset after 5 seconds of inactivity.

The operation of the display shall be as follows:

TABLE V. *Display functions*

Input	Function	Action
Single quick depression	Wake-up	The display shall illuminate the previously displayed field within 2 seconds.
Multiple quick depression	Cycle display	The display shall cycle to the next field with each depression in not less than 0.25 seconds following the order of FIGURE 2.
Long depression	Orientation	The orientation of the display shall rotate 180 degrees. If the display is at rest, the display shall illuminate the previously displayed field within 2 seconds.

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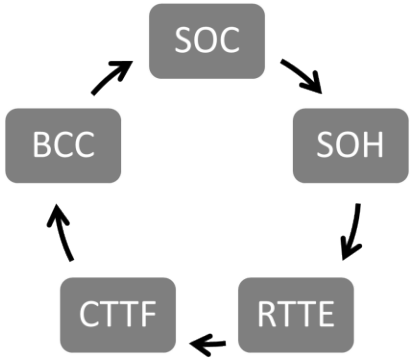


FIGURE 2. Display circular display function transition

e. If the battery is not in a usable condition, the digital display shall flash 000. The display shall continue indefinitely with 000 flashing until the fault is gone or the battery is fully depleted.
f. When illuminated, the output of the display shall be 1.0 to 10.0 millicandelas when measured in a direction normal to the display. The output intensity shall be certified.
g. The push-button shall be capable of operation by Soldiers wearing Mission Oriented Protective Posture (MOPP) IV and cold weather gloves. The push-button actuator shall comply with the following displacement and resistance requirements in TABLE VI.

TABLE VI. *Push button requirements*

Requirement	Not less than	Not greater than
Displacement	0.008 in.	0.050 in.
Resistance	0.25 lbs.	5.50 lbs.

3.5.8.3 Accuracy. The values of the display shall be accurate within +0/-5% of the actual values for the battery. See 4.7.2.14.3.

3.5.9 Communication protocols.

3.5.9.1 SMBus. When specified (see 3.1), batteries shall be compliant with System Management Bus (SMBus) Specification Revision 1.1 and Smart Battery Data (SBData) Specification, version 1.1, with the exception that SBData safety signal hardware requirements therein shall be replaced with a charge enable when a charge enable is specified (see 3.1 and 3.5.6). Certification is required. Batteries shall be compatible with appropriate Level 2 and Level 3 chargers (see 6.4.7). When tested as specified in 4.7.2.15.1, SMBus data output shall be accurate within +0/-5% of the actual state of charge for the battery under test throughout the discharge. Manufacturer and battery data shall be correctly programmed (see 4.7.2.15.1).

SMBus shall use 5V logic circuitry. Pull-up resistors will be provided by the charger. SMBus circuitry shall respond to a SMBus query within 2 seconds. Smart batteries may act as master or

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slave on the bus, but must perform bus master timing arbitration according to the SMBus specification when acting as master.

3.5.9.2 USB power delivery. When specified (see 3.1), batteries shall be compliant with USB PD Specification. When tested as specified in 4.7.2.15.2, batteries shall pass all USB PD 3.1 compliance tests.

3.5.9.2.1 Dual-role power device. When specified (see 3.1), batteries shall be able to utilize the USB as a source, acting as a dual-role power device as specified in the USB PD 3.1 specification. When tested as specified in 4.7.2.15.2, batteries shall advertise source power data objects in its Source_Capabilities message.

3.5.10 Normal operating temperatures. Batteries shall be capable of operating over a temperature range of -4°F to 122°F (-20°C to 50°C). When tested as specified in 4.7.2.7 and 4.7.2.8 batteries shall meet the capacity requirements of 3.5.3 and the visual mechanical requirements of TABLE XIV.

3.5.10.1 Charge acceptance. When tested as specified in 4.7.2.9, batteries shall meet the capacity requirements 3.5.3 and the visual mechanical requirements of TABLE XIV. The surface temperature of the battery shall not exceed 185°F (85°C).

3.5.10.2 Retention of charge. Batteries shall be capable of maintaining the required minimum capacity after storage at 122°F (50°C) for 7 days. When tested as specified in 4.7.2.10, batteries shall meet the capacity requirements 3.5.3 and the visual mechanical requirements of TABLE XIV.

3.5.11 Concurrent charge/discharge. When specified (see 3.1), the battery shall be able to concurrently charge and discharge. When tested as specified in 4.7.2.16, the battery shall be capable of being simultaneously charged and discharged as specified (see 3.1).

3.5.12 Additional electrical requirements. When specified (see 3.1), batteries shall be subjected to additional electrical requirements.

3.6 Environmental requirements. Each battery when subjected to specified environmental tests of TABLE VII below shall show no:

a. Defects listed in TABLE XIV
b. Dimensional distortion beyond specified limits (see 3.1).
c. Sharp current or over voltage fluctuations during any charge or discharge period.
d. Mechanical failure of any part.
e. Failure to mate with a corresponding part.
f. Diminution of the rated capacity below the minimum specified in the applicable specification sheet following the performance of each test.
g. Electrical leakage.
h. Evidence of rupturing, burning or exploding.

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i. Loss of functionality.

TABLE VII. *Environmental requirements*

Tests	Requirement Paragraph	Test Paragraph
Extreme low temperature discharge	3.6.1	4.7.3.2
Extreme high temperature discharge	3.6.1	4.7.3.3
Altitude	3.6.2	4.7.3.4
Temperature shock	3.6.3	4.7.3.5
Mechanical shock	3.6.4	4.7.3.6
Vibration	3.6.5	4.7.3.7
Immersion	3.6.6	4.7.3.8
Transit drop	3.6.7	4.7.3.9
Battery storage life	3.6.8	4.7.3.10
Impact	3.6.9	4.7.3.11
Solar radiation	3.6.10	4.7.3.12
Salt fog	3.6.11	4.7.3.13
Electrostatic discharge	3.6.12	4.7.3.14
Chemical resistance	3.6.13	4.7.3.15
Rapid decompression	3.6.14	4.7.3.16
Explosive atmosphere	3.6.15	4.7.3.17
Combined temperature, altitude, and humidity	3.6.16	4.7.3.18
Additional environmental requirements	3.6.17	4.7.3.19

3.6.1 Extreme discharging temperatures. Type III (Li-ion) batteries shall be capable of discharging over a temperature range of -22°F to 131°F (-30°C to 55°C). When tested as specified in 4.7.3.2 and 4.7.3.3 the batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.2 Altitude. Batteries shall experience no deleterious effects from high altitude. When tested as specified in 4.7.3.4, batteries: (1) shall meet the requirements of 3.6; (2) batteries in the fully charged state shall meet the battery voltage requirements of 3.5.2.

3.6.3 Temperature shock. Batteries shall not bulge, crack, break, leak electrolyte, rupture, burn, or explode as a result of varying rates of thermal expansion or contraction of cell and battery components when subjected to temperature shock temperature cycling in the range of 167°F to -75°F (75°C to -59°C). When tested as specified in 4.7.3.5, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.4 Mechanical shock. Batteries shall be capable of withstanding mechanical shock environments without sustaining physical or electrical damage. When tested as specified in 4.7.3.6, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

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3.6.5 Vibration. Batteries shall be capable of withstanding vibration environments without sustaining physical or electrical damage. When tested as specified in 4.7.3.7 and 4.7.3.7.1: (1) batteries shall exhibit no voltage fluctuations during vibration; (2) shall meet the requirements of 3.6; (3) shall meet the capacity requirements of 3.5.3.

3.6.6 Immersion. Batteries shall be capable of withstanding water immersion without sustaining physical or electrical damage. When tested as specified in 4.7.3.8.1 or 4.7.3.8.2, batteries shall: (1) show no evidence of leakage of water into cases; (2) exhibit weight gain of not greater than 0.01 oz (0.3 g); (3) shall meet the requirements of 3.6.

3.6.7 Transit drop. Batteries shall be capable of withstanding drops at severe temperature conditions without sustaining physical or electrical damage. When tested as specified in 4.7.3.9.1 or 4.7.3.9.2, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6 (with the exception of defects in 106 of TABLE XIV, and minor reworkable deformations beyond specification limits).

3.6.8 Battery storage life. Each battery shall be capable of not less than 3 years of warehouse storage, without any maintenance during storage, during which storage temperatures may vary between the limits of -4°F to 122°F (-20°C to 50°C). When tested as specified in 4.7.3.10, batteries shall, after 3 full cycles of charge and discharge, meet the capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.8.1 Battery sleep mode. Unless otherwise specified, batteries may enter manufacturer specific sleep mode (see 6.4.10) during storage.

3.6.8.2 Battery wake up. If sleep mode is utilized (see 3.6.8.1), unless otherwise specified, batteries shall become fully operational immediately following any battery activity (i.e. button press, charge current, or battery load). Certification is required.

3.6.9 Impact. When specified (see 3.1), batteries shall be capable of being impacted without sustaining physical or electrical damage. When tested as specified in 4.7.3.11 batteries shall meet the impact test capacity requirements of 3.5.3 and the requirements of 3.6 (with the exception of defects in 106 of TABLE XIV).

3.6.10 Solar radiation. When specified (see 3.1), batteries shall be capable of withstanding prolonged exposure to sunlight without sustaining physical or electrical damage. When tested as specified in 4.7.3.12 batteries shall meet the requirements of 3.5.3 and the requirements of 3.6.

3.6.11 Salt fog. When specified (see 3.1), batteries shall withstand a salt fog atmosphere. When tested as specified in 4.7.3.13 batteries for unconnected, unplugged/unprotected power connector shall meet the salt fog test capacity requirements of 3.5.3 and the requirements of 3.6.

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3.6.12 Electrostatic discharge (ESD). When specified (see 3.1), batteries shall withstand ESD to ±8 kV, contact discharge and ±15 kV air discharge on any surface, panel component (including vent), charge contacts, or power / data connectors, including exposed pins, in both operating and non-operating mode. When tested as specified in 4.7.3.14 batteries shall meet the ESD capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.13 Chemical resistance. When specified (see 3.1), batteries shall not be affected by common chemicals in the operational environment such as military decontamination or fuel. When testing in accordance of 4.7.3.15 batteries shall meet the requirements of 3.6.

3.6.14 Rapid decompression. When specified (see 3.1), batteries shall not bulge, crack, break, leak electrolyte, rupture, burn, or explode as a result of rapid decompression. When tested as specified in 4.7.3.16 and 4.7.3.5, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.15 Explosive atmosphere. When specified (see 3.1), batteries shall operate in fuel/air explosive atmospheres without causing ignition. When tested as specified in 4.7.3.17 and 4.7.3.5, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.16 Combined temperature, altitude, and humidity. When specified (see 3.1), batteries shall not bulge, crack, break, leak electrolyte, rupture, burn, or explode as a result of the combined temperature, altitude, and humidity. When tested as specified in 4.7.3.5 and 4.7.3.18, batteries shall meet the capacity requirements of 3.5.3 and the requirements of 3.6.

3.6.17 Additional environmental requirements. When specified (see 3.1), batteries shall be subjected to additional environmental requirements.

3.7 Safety. All protective devices employed shall not adversely affect battery performance or life. Batteries shall be evaluated for any unsafe conditions. An unsafe condition includes but is not limited to, (a) temperature above 338°F (170°C) on the surface of the battery case, (b) sustained flame and/or smoke generation, (c) violent reaction or (d) ejection of materials that present an unsafe condition to the user.

3.7.1 Cell safety.

3.7.1.1 Cell electrical safety. Each cell when subject to the specified electrical tests of TABLE VIII shall not:

a. Explode
b. Catch Fire
c. Rupture
d. Spark
e. Leak any material
f. Have a loss of mass greater than 0.1%
g. Have a surface temperature that exceeds 200°F (93°C)

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TABLE VIII. *Cell electrical safety tests*

Tests	Req't Paragraph	Test Paragraph
Cell overcharge	3.7.1.1.1	4.7.4.1
Cell short circuit	3.7.1.1.2	4.7.4.2
Cell forced discharge	3.7.1.1.3	4.7.4.3

3.7.1.1.1 Cell overcharge. Cells used to build batteries shall be capable of withstanding overcharge (see 4.7.4.1).

3.7.1.1.2 Cell short circuit. Cells used to build batteries shall be capable of withstanding short circuiting (see 4.7.4.2).

3.7.1.1.3 Cell forced discharge. Cells used to build batteries shall be capable of withstanding forced discharge (see 4.7.4.3).

3.7.1.2 Nail penetration, cell. When specified (see 3.1), cells shall not react violently when cell containers are penetrated by sharp objects. Cells tested in accordance with 4.7.4.4 shall not burn or explode, and the external temperature of each test sample shall be not greater than 338°F (170ºC). The results of this test shall be evaluated and shall be not greater than a MEDIUM risk in accordance with MIL-STD-882.

3.7.1.3 Crush, cell. When specified (see 3.1), cells shall not react violently when crushed. When subjected to the test of 4.7.4.5, no cell shall burn or explode.

3.7.2 Battery safety

3.7.2.1 Insulation resistance. All internal insulation used shall have an insulation resistance of not less than 25 megaohms when tested as specified in 4.7.4.6.

3.7.2.2 Overcharge/electrical leakage. Batteries shall be capable of withstanding overcharge conditions without failure. When tested as specified in 4.7.4.7: batteries shall not exceed the voltage limit specified following overcharge (see 3.5.2); no potential in excess of 0.5 V shall be obtained from the case or non-connected terminal to any terminal; and batteries shall meet the visual mechanical requirements of TABLE XIV and 3.6, and shall meet the capacity requirements of 3.5.3. The surface temperature of the battery shall not exceed 185°F (85°C).

3.7.2.3 Short circuit protection. Each battery shall be protected against short circuits. When tested as specified in 4.7.4.8, there shall not be any damage to the battery, and the battery shall be able to meet the full discharge capacity requirement of 3.5.3 after full charge.

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3.7.2.4 High temperature temporary cut off devices. Each battery shall contain a minimum quantity of normally closed thermoswitches that shall open at 158 ± 9°F (70 ± 5°C) and close at 122 ± 9°F (50 ± 5°C). Each thermoswitch shall make physical contact with not less than one cell. The minimum quantity shall be as specified (see 3.1). The quantity of thermoswitches shall be certified. When tested as specified in 4.7.4.9, battery voltage shall be zero volts after each high temperature storage and batteries shall meet the voltage requirement of 3.5.2 after cooling. After completion of the test the battery shall be able to meet the full discharge capacity requirement of 3.5.3 after full charge.

3.7.2.5 High temperature permanent cut off devices. Each battery shall be permanently shut off when the temperature of the battery reaches 199 ± 9°F (93 ± 5°C). The device shall prevent charging and discharging of the battery. The minimum quantity shall be as specified in the applicable specification sheet. When tested as specified in 4.7.4.10, battery voltage shall be zero volts after high temperature storage and shall remain at zero after 104°F (40°C) storage.

3.7.2.6 Inter-cell protective devices. When specified (see 3.1), batteries shall include protective devices/circuitry/material to prevent thermal event propagation between cells. Certification is required.

3.7.2.7 Redundant safety protection. When specified (see 3.1), batteries shall include the specified redundant protective devices.

3.7.2.8 Projectile. When specified (see 3.1), batteries shall not present a risk to the user when subjected to a flame. When tested as specified in 4.7.4.11, the production of smoke and flames and the passing of particles (smaller than mesh openings) through the wire mesh are permitted; heat damage to the wire mesh is also permitted. Test must be set up in accordance to UL1642 Projectile Test. During test, batteries must be secured to the screen with a single wire tied around the sample. Video of the test is required from start of testing to end of testing for each battery. Pictures are to be taken and to include but are not limited to; pre-test apparatus (including all equipment), pre-test battery, post-test battery, post- test apparatus (including all equipment), post-test room, and any post-test shrapnel found with reference object. The results of this test shall be evaluated and shall be not greater than a MEDIUM risk in accordance with MIL-STD-882.

3.7.2.9 Nail penetration, battery. When specified (see 3.1), batteries shall not react violently when battery containers are penetrated by sharp objects. Batteries tested in accordance with 4.7.4.12 shall not burn or explode, and the external temperature of each test sample shall be not greater than 338°F (170°C). The results of this test shall be evaluated and shall be not greater than a MEDIUM risk in accordance with MIL-STD-882.

3.7.2.10 Lithium battery safety program (US Navy). When specified (see 3.1), Types III batteries shall be designed and built in a manner that provides protection from bulging, leaking, venting, rupturing, burning and exploding under conditions of extreme electrical abuse.

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Additionally, batteries shall be designed in a manner that prevents the production of shrapnel in the event of failure of internal protection devices under extreme conditions.

3.7.2.10.1 Destructive lithium battery safety program. When tested in accordance with 4.7.5.1 through 4.7.5.3, the battery case shall not fragment. Case distortion, cracking, or bulging, and venting of gases or liquids through purposely designed cell vents, are permitted.

3.7.2.10.2 Non-destructive lithium battery safety program. When tested in accordance with 4.7.5.4, no battery shall bulge, leak, vent, rupture, burn, or explode.

3.7.2.11 Weapons characterization. When specified (see 3.1), batteries shall be capable of safely withstanding penetration by bullets/projectiles. When tested as specified in 4.7.4.13, batteries shall not have thermal propagation to adjacent cells. Batteries shall not exhibit unsafe conditions of 3.7 for greater than two minutes. The results of this test shall be evaluated and shall be not greater than a MEDIUM risk in accordance with MIL-STD-882.

3.7.3 Additional safety requirements. When specified (see 3.1), batteries shall be subjected to additional safety requirements.

3.8 Electromagnetic interference. When specified (see 3.1), the battery shall be electromagnetically compatible with the system specified in the applicable specification sheet. When tested as specified in 4.7.7, batteries shall meet the specified requirements.

3.9 Transportation requirements. Batteries and cells shall be tested per applicable US DOT and UN Manual of Tests and Criteria shipping regulations and meet requirements therein. Certification is required.

3.9.1 Additional transportation requirements. When specified (see 3.1), batteries shall be subjected to additional transportation requirements.

3.10 Interoperability.

3.10.1 Charger interoperability. Batteries shall be interoperable with currently fielded chargers and adapters as specified (see 3.1). When tested as specified in 4.7.6.1, batteries shall reach not less than 90% SOC and show no anomalies.

3.10.2 End item interoperability. Batteries shall be interoperable with end items as specified (see 3.1). When tested as specified in 4.7.6.2, batteries shall operate the end item as intended and show no anomalies.

3.11 Reliability. When specified (see 3.1), batteries shall meet the specified reliability requirements (see 4.7.2.4.1).

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4 VERIFICATION

4.1 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Conformance inspection (see 4.5).

4.2 Test equipment and facilities.

4.2.1 Electrical indicating instruments. All voltmeters and ammeters shall be accurate within 1% of the full scale reading. The sensitivity of voltmeters shall be not less than 10 megaohms per volt.

4.2.2 Resistor and current tolerances. In all tests involving a resistor, such resistance shall be accurate within ± 0.5% during the charging and discharging of batteries, conducted at various rates specified herein, the current shall be maintained within a tolerance of ± 3% of the specified value.

4.2.3 Timing. The timing of discharges lasting not greater than one minute shall be maintained within ± 5%. All others shall be accurate within ± 1%.

4.3 Inspection conditions. Unless otherwise specified herein, all examinations and tests shall be performed under ambient temperature, humidity, and atmospheric pressure conditions.

4.3.1 Normal conditions. When normal conditions are specified, tests shall be conducted and measurements taken at 73.4 ± 9°F (23 ± 5°C) and ambient atmospheric pressure and relative humidity with not less than 2 hours between charge and discharge except for cycle life test.

4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in TABLE IX and TABLE X. Testing shall be performed in the order shown. All samples shall meet the requirements specified herein for each examination and test. Samples shall be as follows:

a. TABLE IX tests require not greater than ten cells
b. Five pieces of battery cases may be required (see 3.4.5)
c. Two electronic component assemblies (ECA) and two complete cell assemblies (CCA) (see 6.4.1and 6.4.5 for definitions)
d. Group 1 (GR1) – Functional tests, shall contain not less than 12 samples. Additional samples may be required based on test specified (see 3.1).
e. Group 2 (GR2) – Electromagnetic interference tests, when specified (see 3.1) shall contain five samples
f. Group 3 (GR3) – Interoperability tests, shall contain eight samples
g. Group 4 (GR4) – Lithium safety tests, when specified (see 3.1) shall contain 12 samples

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h. Group 5 (GR5) – Nail penetration tests (battery level) and Weapons Characterization tests shall contain the specified number of samples (see 3.1)
i. Group 6 (GR6) – Reliability tests shall contain 10 samples

These quantities may be less for some battery types based on applicability of the tests.

TABLE IX. *First article tests and certifications; cells, assemblies, and components*

Inspection	Requirement Paragraph	Test Method Paragraph	Samples
Cell balance[1]	3.5.1	4.7.2.1	Cells: All
Cell overcharge[1]	3.7.1.1.1	4.7.4.1	Cells: 1, 2
Cell short circuit[1]	3.7.1.1.2	4.7.4.2	Cells: 3, 4
Cell force discharge[1]	3.7.1.1.3	4.7.4.3	Cells: 5, 6
Crush[1,2]	3.7.1.3	4.7.4.5	Cells: 7, 8
Nail penetration[1,2]	3.7.1.2	4.7.4.4	Cells: 9, 10
Power consumption[1]	3.5.7	4.7.2.13	2 ECA & 2 CCA
Insulating compounds, flow and shrinking[1]	3.3.4	4.7.1.5	By volume
Battery case[1]	3.4.5	4.7.1.7	5 pieces or certification
Metals	3.3.1		Certification
Dissimilar metals	3.3.2		Certification
Resistance of elastomeric materials	3.3.5		Certification
Electrical connection wires and tabs	3.3.6		Certification
Electrical contacts	3.4.4.1		Certification
State of charge indicator window	3.4.5.1		Certification
Dust cap or cover	3.4.7.1		Certification
Transportation requirements	3.9		Certification

NOTES

[1] Must be completed prior to tests of TABLE X

[2] Only when specified (see 3.1)

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TABLE X. *First article tests, complete batteries*

Inspection/Test	Req't	Test Method	Samples or Group (GR)															
			GR 1										GR 2	GR 3	GR 4	GR 5		GR 6
			1	2	3	4	5,6	7,8	9, 10	11, 12	13, 14	15, 16	(5 ea)	(8 ea)	(12 ea)	1-6	7 - 18	(10 ea)
Visual and mechanical	3.4	4.7.1.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	☑	✓	☑	☑	☑	☑
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				☑	☑	☑
Battery open circuit voltage	3.5.2	4.7.2.2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	☑	✓	✓	☑	☑	☑
Insulation resistance	3.7.2.1	4.7.4.6	✓	✓	✓	✓	✓	✓	✓	✓								
Altitude	3.6.2	4.7.3.4	✓	✓	✓	✓	✓	✓	✓	✓								
Charge acceptance	3.5.10.1	4.7.2.9										✓						
Concurrent charge/discharge	3.5.11	4.7.2.16										✓						
Capacity discharge (initial)	3.5.3	4.7.2.3.1	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				☑	☑	☑
Cycle life	3.5.4	4.7.2.4					✓											
Battery storage life	3.6.8	4.7.3.10						✓										
Overcharge / electrical leakage	3.7.2.2	4.7.4.7							✓									
Low temperature discharge	3.6.1	4.7.2.7							✓									
High temperature discharge	3.6.1	4.7.2.8							✓									
Projectile	3.7.2.8	4.7.4.11							☑									
Nail Penetration, battery	3.7.2.9	4.7.4.12														☑		
High rate discharge	3.5.3	4.7.2.5	✓		✓													
Retention of charge	3.5.10.2	4.7.2.10	✓	✓	✓	✓												
Pulse discharge	3.5.3	4.7.2.11	✓		✓													
Inrush current	3.5.5	4.7.2.12								☑								
Temperature shock	3.6.3	4.7.3.5	✓	✓	✓	✓												
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓												
Vibration (during test)	3.6.5	4.7.3.7	✓		✓													
Vibration (loose cargo)	3.6.5	4.7.3.7.1		☑		☑												
Chemical resistance	3.6.13	4.7.3.15									☑							
Connector insertion	3.4.4.1.1.2	4.7.1.3	✓			✓												
Flat terminal strength	3.4.4.1.2	4.7.1.4	✓			✓												
USB power port connector(s)	3.4.4.1.5	4.7.2.17	☑	☑	☑	☑												

NOTES

This symbol ☑ indicates only when specified (see 3.1)

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TABLE X. *First article tests, complete batteries* - Continued

Inspection/Test	Req't	Test Method	Samples or Group (GR)															
			GR 1										GR 2	GR 3	GR 4	GR 5		GR 6
			1	2	3	4	5,6	7,8	9, 10	11, 12	13, 14	15, 16	(5 ea)	(8 ea)	(12 ea)	1-6	7 - 18	(10 ea)
Immersion, shallow	3.6.6	4.7.3.8.1	☑	☑	☑	☑						☑						
Immersion, deep	3.6.6	4.7.3.8.2										☑						
Transit drop, normal	3.6.7	4.7.3.9.1	☑	☑	☑	☑												
Transit drop, severe	3.6.7	4.7.3.9.2										☑		☑				
Surface friction	3.4.5.6	4.7.1.8		☑	☑													
Reliability	3.11	4.7.2.4.1																☑
Impact	3.6.9	4.7.3.11									☑							
Salt fog	3.6.11	4.7.3.13								☑								
ESD	3.6.12	4.7.3.14	☑	☑	☑	☑												
Solar radiation	3.6.10	4.7.3.12								☑								
Rapid decompression	3.6.14	4.7.3.16								☑								
Explosive atmosphere	3.6.15	4.7.3.17								☑								
Combined temperature, humidity, and altitude	3.6.16	4.7.3.18									☑							
Immersion, shallow (post drop)	3.6.6	4.7.3.8.1	☑	☑								☑		☑				
Immersion, deep (post drop)	3.6.6	4.7.3.8.2										☑						
State of charge indicator	3.5.8	4.7.2.14									☑							
SMBus	3.5.9.1	4.7.2.15.1	☑	☑	☑	☑												
Full capacity discharge	3.5.3	4.7.2.3	☑	☑	☑	☑												
Concurrent charge / discharge	3.5.11	4.7.2.16	✓			✓												
Extreme low temp. discharge	3.6.1	4.7.3.2		✓		✓												
Extreme high temp. discharge	3.6.1	4.7.3.3	✓		✓													
Battery case vent	3.4.5.3	4.7.1.6										✓						
Short circuit protection	3.7.2.3	4.7.4.8									☑							

NOTES

This symbol ☑ indicates only when specified (see 3.1)

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TABLE X. *First article tests, complete batteries* - Continued

Inspection/Test	Req't	Test Method	Samples or Group (GR)														
			GR 1									GR 2	GR 3	GR 4	GR 5		GR 6
			1	2	3	4	5,6	7,8	9, 10	11, 12	13, 14	15, 16	(5 ea)	(8 ea)	(12 ea)	1-6	7 - 18	(10 ea)
High temp. temporary cut-off	3.7.2.4	4.7.4.9		✓	✓													
High temp. permanent cut-off	3.7.2.5	4.7.4.10		✓	✓													
Electromagnetic interference	3.8	4.7.7											☑					
End item interoperability	3.10.2	4.7.6.2												☑				
Charger interoperability	3.10.1	4.7.6.1												✓				
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5													☑			
Weapons characterization	3.7.2.11	4.7.4.13															☑	

NOTES
This symbol ☑ indicates only when specified (see 3.1)

4.5 Conformance inspection. Inspection of product for delivery shall consist of groups A and B inspections, as well as Group C inspections when specified (see 4.5.3.1).

4.5.1 Group A inspection. Each battery on contract or delivery order shall be 100% inspected for conformance to the inspections in the order specified in TABLE XI. All failures shall be removed. In the event of failure, rework is permitted only as specified in TABLE XIV. Reworked batteries must be fully compliant after rework.

TABLE XI. *Group A inspection*

Inspection	Requirement Paragraph	Test Method paragraph
Visual and mechanical examination	3.4	4.7.1.1
Battery open circuit voltage	3.5.2	4.7.2.2
Capacity discharge (initial)[1]	3.5.3	4.7.2.3.1
Immersion, shallow (modified)[1]	3.6.6	4.7.3.8.1

NOTES
[1] When specified (see 3.1)

4.5.2 Group B inspection. Group B inspections shall be performed on inspection lots that have passed Group A inspection (see 4.5.1) and on samples selected from units that have been subjected to and met the Group A inspection. Group B samples selection shall be performed on each inspection lot (see 6.8). Group B inspection shall consist of the test specified in TABLE XII and shall be conducted in the order shown. The sample size shall be no less than eight

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batteries for each inspection lot (see 6.8) or each month of production if an inspection lot spans more than one month. Group B samples may not be provided as deliverable items unless specified otherwise by the contract.

TABLE XII. *Group B inspection*

Inspection / Criteria	Requirement Paragraph	Test Method Paragraph	Group B Sample Percentage			
			GR-B1 25%	GR-B2 25%	GR-B3 25%	GR-B4 25%
High rate discharge	3.5.3	4.7.2.5	✓	✓	✓	✓
Concurrent charge / discharge 1/	3.5.11	4.7.2.16	✓	✓	✓	✓
SMBus	3.5.9.1	4.7.2.15.1	✓	✓	✓	✓
Immersion, shallow	3.6.6	4.7.3.8.1	✓	✓	✓	✓
Transit drop, severe 1/	3.6.7	4.7.3.9.2	✓	✓		
ESD 1/	3.6.12	4.7.3.14	✓	✓	✓	✓
Temperature shock	3.6.3	4.7.3.5	✓	✓		
Capacity discharge (initial)	3.5.3	4.7.2.3.1			✓	✓

NOTES
1/ when specified (see 3.1)

4.5.3 Group C inspection. Group C inspections shall be performed on inspection lots that have passed Group A inspection (see 4.5.1) and on samples selected from units that have been subjected to and met the Group A inspection. This inspection is required during conformance inspection for the first inspection lot and repeated in accordance with 4.5.3.1. It shall consist of the tests listed in TABLE XIII. Inspection shall be performed in the order shown. The minimum battery sample size shall be not less than eight batteries and eight cells. Group C samples may not be provided as deliverable items unless specified otherwise by the Government.

4.5.3.1 Group C frequency. Group C testing, or a complete first article inspection (see 4.4), shall be performed within the 12 months prior to presenting the current inspection lot for acceptance. Each group shall consist of two samples.

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TABLE XIII. *Group C inspection*

Inspection / Criteria	Requirement Paragraph	Test Method Paragraph	Group C Sample			
			GR-C1	GR-C2	GR-C3	GR-C4
Dimensions and weight	3.4.3	4.7.1.2	✓	✓	✓	✓
Capacity discharge (initial)	3.5.3	4.7.2.3	✓	✓	✓	✓
Impact	3.6.9	4.7.3.11	☑	☑		
Salt fog	3.6.11	4.7.3.13		✓		✓
Overcharge/electrical leakage	3.7.2.2	4.7.4.7	✓		✓	
High temperature discharge	3.6.1	4.7.2.8	✓	✓		
Low temperature discharge	3.6.1	4.7.2.7			✓	✓
Temperature shock	3.6.3	4.7.3.5	✓	✓	✓	✓
Mechanical shock	3.6.4	4.7.3.6	✓	✓	✓	✓
Vibration	3.6.5	4.7.3.7	✓	✓	✓	✓
Immersion, shallow	3.6.6	4.7.3.8.1	✓	✓	✓	✓
Transit drop, normal	3.6.7	4.7.3.9.1	✓	✓	✓	✓
Immersion, shallow (post drop)	3.6.6	4.7.3.8.1	✓	✓	✓	✓
Capacity discharge (final)	3.5.3	4.7.2.3	✓	✓	✓	✓
Extreme low temp. discharge	3.6.1	4.7.3.2	☑			☑
Extreme high temp. discharge	3.6.1	4.7.3.3	☑			☑
Battery case vent	3.4.5.3	4.7.1.6	✓			✓
High temperature permanent cut-off	3.7.2.5	4.7.4.10		✓	✓	
Nail penetration, battery	3.7.2.9	4.7.4.12	✓	✓	✓	✓
Cell balance	3.5.1	4.7.2.1	8 cells			

NOTES

This symbol ☑ indicates only when specified (see 3.1)

4.6 Charge methods.

4.6.1 Constant potential, current limited. Where specified herein, Type III (Li-ion) batteries shall be charged by a constant potential (voltage) method with limited current at normal conditions in accordance with 4.3.1 at a rate not less than specified (see 3.1) and for not greater than the specified time (see 3.1). Voltages, current limits, and cut-off parameters are specified in the applicable specification sheet (see 3.1).

4.6.2 Constant current. Where specified herein, Type I (Ni-Cd) and Type II (Ni-MH) batteries shall be charged by a constant current to the designated charge cut-off parameter. Topping charges beyond the charge cut-off parameter shall be as specified. Charge and topping charge current and cut-off parameters are specified in the applicable specification sheet (see 3.1).

4.6.3 Alternate charging.

a. Where permitted herein, batteries may be charged by an approved charger as an alternative to the charge methods described above (see 6.6).

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b. If an approved charger is not available, the alternate charging method described by the applicable specification sheet may be used whenever charging in accordance with 4.6.3 is permitted.

4.7 Test methods. Not all test methods are applicable to all batteries. See section 3 requirements to determine if applicable.

4.7.1 Physical characteristics.

4.7.1.1 Visual and mechanical. Samples shall be examined to verify that the basic materials, component materials and parts, design and construction, marking and workmanship are in accordance with all the requirements of 3.4 and TABLE XIV.

TABLE XIV. *Visual and mechanical inspection*

Categories 1/	Rework Permitted	Defects
001	No	Electrolyte leakage.
002	No	Improper assembly causing parts to be inoperative or unsafe in service.
003	No	Loose parts inside battery.
101	No	Deformed or damaged parts which are inoperative or malfunction in service
102	Yes	Contact surfaces obstructed so that electrical use is affected. Rust, corrosion, or any type of non-conductive material on contact surfaces.
103	No	Battery cases – any hole (not specified), tear, rip, or crack.
104	Yes	Improper case closure.
105	Yes	Sharp edges on exterior of battery.
106	No	Pitting, blow-holes, rough spots, burrs, splits, dents, or other deformations on battery surface.
107	Yes	Terminal, identification, display, or operating instructions markings not as specified (see 3.1 and 3.4.6).
108	Yes	Marking defects of 3.4.6.8 are visible
109	Yes	State of charge indicator does not display full state of charge (Group A only).
110	No	State of charge indicator does not display correct state of charge.
111	No	State of charge indicator window is warped, punctured, separated from case, not transparent, or has visible condensation on the interior.
201	No	Color not as specified.

NOTES
1/ Category 0XX defects are critical; category 1XX are major, and category 2XX are minor. These categories are used to qualify the levels of nonconformance. Critical defects affect safety; major categories affect use, batteries with minor defects may be serviceable with limitations.

4.7.1.2 Dimensions and weight. Batteries shall be examined by gauging or measuring and by weighing to determine conformance. All measurements shall include any coating and labels which may be used. Battery dimensions shall be measured during the dimension inspections of

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first article. Dimensions may be checked by gauging whenever a first article test requires dimensional compliance as passing criteria following tests. Dimension inspections during conformance inspections may be accomplished by gauging. Batteries shall meet the requirements specified (see 3.1).

4.7.1.2.1 Gauging. When box or ring gauges are used, batteries, loaded with a weight not greater than five pounds, shall pass freely through the applicable gauge openings. The dimensions of the box or ring gauge shall be the specified maximum outside dimensions of the battery (see 3.1).

4.7.1.3 Connector insertion test. Unless otherwise specified (see 3.1), the following test shall be performed. Insert and withdraw each battery into a specified adapter or specified mating device (see 3.1) 500 times. Measure insertion and withdrawal force required on the first and last insertion/withdrawal cycles. Inspect the battery to verify that the screws (if any) are retained, the connector remains securely fastened and the connector remains fully functional. Batteries shall meet the requirements of 3.4.1 and 3.4.4.1.1.2.

4.7.1.4 Flat terminal strength. Unless otherwise specified (see 3.1), the following test shall be performed. Apply force of not less than 10 pounds to each battery terminal using 0.200 ± 0.050 inch diameter rod. Repeat force application for not less than a total of 250 applications. Maintain each application of force for not less than 1 second. Batteries shall meet the requirements of 3.4.4.1.2.

4.7.1.5 Insulating compounds, flow and shrinking. The following test shall be performed on any insulating compounds that may be used in battery construction. Place compound in a container, approximately 3 inches wide by 6 inches long by 3/4 inch high, to within 1/2 inch of the top and allow curing in accordance with the manufacturer's recommended procedures. Place the container in an environmental chamber in an inverted position and raise the temperature to 200 ± 4.8°F (93.3 ± 2.7ºC). Keep the container inverted at these conditions for not less than 24 hours. Visually examine; the test sample shall meet the requirements of 3.3.4. Place the container in an environmental chamber and lower the temperature to -40 ± 4.8°F (-40 ± 2.7ºC); hold these conditions for not less than eight hours. Visually examine; the test sample shall meet the requirements of 3.3.4.

4.7.1.6 Battery case vent. The following test shall be performed. Place a fully assembled battery into a fixture designed to contain possible bulging by the front, back and side faces of the battery case (see applicable specification sheet for face references). The fixture shall allow the battery to expand up to .010 inches over the specified maximum allowable dimensions. The fixture shall not cover the top and bottom of the battery. Tap an air line into the battery and increase the internal pressure of the battery 1.0 psi every 60 seconds until the battery case vent opens. Position the fixture with the pressurized battery so that it allows the battery to slide out through the bottom of the fixture. If the pressurized battery cannot slide out of the fixture, place weights on top of the battery. Measure and record the total weight used including the battery. Visually examine. Batteries shall meet the requirements of 3.4.5.3.

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4.7.1.7 Battery case. When specified (see 3.1), the following test shall be performed on 5 sample pieces of case material. Use the Horizontal Burning Test method of UL 94 to determine the test apparatus criteria. Expose samples to a flame for not less than 30 seconds. Samples shall meet the criteria of 3.4.5.

4.7.1.8 Surface friction. When specified (see 3.1), the following test shall be performed. Verify battery surface is dry and free of any lubricants prior to test. Measure and report the coefficient of static friction per test method D2394 of ASTM G115-10, Standard Guide for Measuring and Reporting Friction Coefficients. Verify that the value is not greater than 0.4. Batteries shall meet requirements in 3.4.5.6.

4.7.1.9 Additional material tests. When specified (see 3.1), batteries shall be subjected to additional specified material tests.

4.7.1.10 Additional physical characteristics. When specified (see 3.1), batteries shall be subjected to additional specified physical characteristics.

4.7.2 Electrical performance tests.

4.7.2.1 Cell balance. The following test shall be performed. Fully charge the cells in any appropriate manner at the normal conditions of 4.3.1. Stabilize cells at normal conditions after charge termination for not less than 1 hour. Discharge cells at the C/5 rate to the cell manufacturer's recommended discharge cutoff voltage. Measure and record total capacity of each cell to cutoff voltage. Cells shall meet the requirements of 3.5.1.

4.7.2.2 Battery open circuit voltage. The open circuit voltage of the battery under test shall be measured. In the case of multi-sectioned batteries, open circuit voltage shall be measured in both series and independent modes (see 3.1). Batteries shall meet the requirements of 3.5.2.

4.7.2.3 Capacity discharge. The following test shall be performed:

a. Charge batteries in accordance with 4.6; use of 4.6.3 is not permitted unless otherwise specified (see 3.1).
b. Stabilize the batteries at the conditions of 4.3.1 for not less than 2 hours.
c. Discharge at the rate specified; batteries shall meet full capacity discharge test requirement (see 3.1 and 3.5.3).
d. When a display is present (see 3.5.8), observe the display and verify operability.

4.7.2.3.1 Initial capacity discharge. Each battery subjected to the capacity discharge test above (see 4.7.2.3) on its initial charge/discharge cycle is permitted up to three cycles to meet the capacity discharge test requirement (see 3.1). Any battery not meeting the specified capacity discharge requirement (see 3.1) during any of its first three cycles is considered a failure.

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4.7.2.4 Cycle life test. The following test shall be performed. Details for each of the phases of a cycle (see 6.4.3) are as follows:

a. Cycles 1 through 27 - Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Allow batteries to rest for not less than five minutes and not greater than one hour. Discharge at the rate specified to the specified cutoff voltage (see 3.1). Allow batteries to rest for not less than five minutes and not greater than one hour prior to the start of the next cycle. During cycles 7, 14, 21, and 27 both rest periods shall be not less than five minutes and not greater than five hours.
b. Cycle 28 - Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Perform the electric leakage test of 4.7.4.7d. Allow batteries to rest for not less than four hours and not greater than 16 hours. Discharge at the rate specified to the specified cutoff voltage (see 3.1). Record battery capacity and allow batteries to rest for not less than five minutes and not greater than one hour. If the battery has a smart display (see 3.5.8.2), verify the BCC and SOH fields are accurate.
c. Repeat steps a through b above until the total number of required cycles is obtained (see 3.1).
d. After completion of the cycle life test, visually examine batteries for the criteria of TABLE XIV. Batteries shall meet the cycle life and visual mechanical requirements (see 3.1 and 3.5.3).

4.7.2.4.1 Reliability test. When specified (see 3.1) the following test shall be performed. Batteries shall undergo cycles 1, 2, 3 and 4 described herein and accrue a cumulative minimum of the specified hours of operation (see 3.1) with zero reliability failure allowed. In addition, batteries shall complete Cycles 1, 2, 3, and 4 sequentially and accrue a minimum of the specified hours each (see 3.1). Record the BCC for the batteries prior to conducting this test.

1. Cycle 1: Battery Operation at 14°F (-10°C).

a. Set chamber temperature to not greater than 14°F (-10°C).
b. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
c. Allow batteries to rest for one hour.
d. Discharge at specified rate (see 3.1) to cutoff voltage.
e. Allow batteries to rest for one hour.
f. Record elapsed time from step b to step e. Record cumulative hours.
g. Repeat Step b to Step f for 3 times.
h. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
i. Allow batteries to rest for one hour.
j. Discharge at specified rate (see 3.1) to cutoff voltage.
k. Allow batteries to rest for one hour.
l. Record elapsed time from step h to step k. Record cumulative hours.
m. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
n. Allow batteries to rest for one hour.
o. Discharge at specified rate (see 3.1) to cutoff voltage.
p. Allow batteries to rest for one hour.

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q. Record elapsed time from step m to step p. Record cumulative hours of Cycle 1.

2. Cycle 2: Battery Operation at 72 ± 6.8°F (22 ± 4°C).

a. Set chamber temperature to 72 ± 6.8°F (22 ± 4°C).
b. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
c. Allow batteries to rest for one hour.
d. Discharge at specified rate (see 3.1) to cutoff voltage.
e. Allow batteries to rest for one hour.
f. Record elapsed time from step b to step e. Record cumulative hours.
g. Repeat Step b to Step f for 10 times.
h. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
i. Allow batteries to rest for one hour.
j. Discharge at specified rate (see 3.1) to cutoff voltage.
k. Allow batteries to rest for one hour.
l. Record elapsed time from step h to step k. Record cumulative hours.
m. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
n. Allow batteries to rest for one hour.
o. Discharge at specified rate (see 3.1) to cutoff voltage.
p. Allow batteries to rest for one hour.
q. Record elapsed time from step m to step p. Record cumulative hours of Cycle 2.

3. Cycle 3: Battery Operation at 104°F (40°C).

a. Set chamber temperature to not less than 104°F (40°C).
b. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
c. Allow batteries to rest for one hour.
d. Discharge at specified rate (see 3.1) to cutoff voltage.
e. Allow batteries to rest for one hour.
f. Record elapsed time from step b to step e. Record cumulative hours.
g. Repeat Step b to Step f for 3 times.
h. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
i. Allow batteries to rest for one hour.
j. Discharge at specified rate (see 3.1) to cutoff voltage.
k. Allow batteries to rest for one hour.
l. Record elapsed time from step h to step k. Record cumulative hours.
m. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
n. Allow batteries to rest for one hour.
o. Discharge at specified rate (see 3.1) to cutoff voltage.
p. Allow batteries to rest for one hour.
q. Record elapsed time from step m to step p. Record cumulative hours of Cycle 3.

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4. Cycle 4: Conduct Battery Essential Function Tests.

a. Charge batteries in accordance to with 4.6; use of 4.6.3 is permitted.
b. Perform the electric leakage test of 4.7.4.7e.
c. Allow batteries to rest for not less 4 hours.
d. Discharge at specified rate (see 3.1) to cut off voltage.
e. Record Battery capacity.
f. Record elapsed time from step b to step e.
g. Record cumulative hours of Cycle 4.

After completion of Cycle 1, 2, 3, and 4 tests, visually examine batteries for the criteria of 4.7.1.1 TABLE XIV. Record the Battery Cycle Count for the batteries.

The total cumulative hours of each battery shall be the cumulative hours of Cycles 1 through 4. Verify that the cumulative hours meet or exceed the requirements of 3.11.

Verify the batteries operate and pass the capacity requirements of 3.6.

4.7.2.5 High rate discharge. The following test shall be performed.

a. Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Stabilize the batteries at ambient temperature for not less than 2 hours.
b. Batteries shall be not less than 40 inches apart and the surface temperatures shall be monitored.
c. Discharge the batteries at the rate specified (see 3.1).
d. Record the following:
    1. Battery open circuit voltage
    2. The elapsed time for the voltage to drop to the cutoff voltage specified on the applicable specification sheet after the start of discharge.
e. After completion of the test, batteries shall meet the capacity requirements of 3.5.3 and the criteria of TABLE XIV.

4.7.2.6 Charge enable. When specified (see 3.1), the following test shall be performed. Discharge each battery to cutoff voltage. Using an ammeter in the charge circuit to record resultant current slowly increase the charge current until the battery no longer accepts a charge current. Once the charge current has been verified, discharge each battery to cutoff voltage. Apply a charge current to the battery terminals while connected to the charge enable terminal or socket using a charge circuit equipped to provide the specified preferred charge method only if the charge enable terminal or socket complies with requirements (see 3.1). After testing, batteries shall meet the requirements of 3.5.6.

4.7.2.7 Low temperature discharge. The following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is not permitted. Store the batteries at -4 ± 2°F (-20 ± 1.1°C) for not less than 4 hours. Discharge at these conditions at the rate specified to the specified cutoff voltage (see 3.1). After testing, batteries shall meet the requirements of 3.5.3 and 3.6.

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4.7.2.8 High temperature discharge. The following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is not permitted. Batteries shall be not less than 40 inches apart and the surface temperatures shall be monitored. Store the batteries at 122 ± 2°F (50 ± 1.1°C) for not less than 4 hours. Discharge at these conditions at the rate specified to the specified cutoff voltage (see 3.1). After testing, batteries shall meet the requirements of 3.5.3 and 3.6.

4.7.2.9 Charge acceptance. The following test shall be performed. After testing, batteries shall meet the requirements of 3.5.10.1.

a. Discharge batteries in accordance with 4.7.2.3.
b. Store the batteries at -4 ± 2°F (-20 ± 1.1°C) for not less than 4 hours.
c. Charge batteries in accordance with 4.6 at the same condition; use of 4.6.3 is not permitted unless otherwise specified (see 3.1).
d. Stabilize batteries at the same condition for not less than 2 hours.
e. Discharge batteries in accordance with 4.7.2.3 at the same condition.
f. Store the batteries at 122 ± 2°F (50 ± 1.1°C), unless otherwise specified (see 3.1), for not less than 4 hours.
g. Repeat steps c through e above.

4.7.2.10 Retention of charge (at high temperature). The following test shall be performed. After testing, batteries shall meet the requirements of 3.5.3 and 3.5.10.2.

a. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
b. Store batteries at 122 ± 2°F (50 ± 1.1°C) for 7 days.
c. Return batteries to the normal conditions of 4.3.1 and stabilize for not less than 2 hours.
d. Discharge batteries at the same conditions at the rate specified to the specified cutoff voltage (see 3.1).

4.7.2.11 Pulse discharge. The following test shall be performed. After testing, batteries shall meet the requirements of 3.5.3.

a. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
b. Batteries shall be not less than 40 inches apart and the surface temperatures shall be monitored.
c. Store batteries under the normal conditions of 4.3.1.
d. Discharge each battery at the rate and for the time specified until the cutoff voltage specified is reached (see 3.1).

4.7.2.12 Inrush current. When specified (see 3.1), the following test shall be performed. During and after testing batteries shall meet the surge requirement (see 3.5.5).

a. Charge the batteries in accordance with 4.6; use of 4.6.3 is permitted. Stabilize the batteries at ambient temperature for not less than two hours.
b. Batteries shall be not less than 40 inches apart and the surface temperatures shall be monitored.
c. Store batteries under the normal conditions of 4.3.1 .

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d. Discharge the batteries using current values not less than the profile specified (see 3.1)
e. Record the following:
    1. Discharge current
    2. Battery Voltage
f. After completion of the test, visually examine batteries for the criteria of TABLE XIV. Batteries shall meet the surge and visual mechanical requirements (see 3.5.5).

4.7.2.13 Power consumption. The following test shall be performed. Record the current between the ECA for the battery and the full CCA for the battery over a one hour period or under specified conditions (see 3.1). Verify the average current draw is in accordance with paragraph 3.5.7.

4.7.2.14 Display. When specified (see 3.1), the following tests shall be performed.

4.7.2.14.1 Display, non-smart. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Verify all five segments of the display are visible.

4.7.2.14.2 Display, smart.

a. Charge each battery in accordance with 4.6; use of 4.6.3 is permitted.
b. Press and release the display button.
c. Measure and record the following to the nearest tenth of a second: (1) time from activation to full intensity of the display; (2) time from reaching full intensity until the lights extinguish.
d. Measure displacement and resistance values for the button and record the results.
e. Verify that the display function's default settings are as specified (see 3.1).
f. Press the display and actuator button face against a flat, transparent surface with not less than 5.50 lbs. of force. Verify the display does not illuminate.
g. While the display is inactive, press and hold the actuator button for greater than 6 seconds. Verify that the display changes its default orientation.
h. While display is still active, press and hold actuator button for greater than 6 seconds. Verify that the display changes back to its default orientation.
i. Press the actuator button with multiple quick depressions (≤ 6 seconds each). Verify that the display cycles in accordance with TABLE V and FIGURE 2. Verify that the display content is in accordance with TABLE IV.

4.7.2.14.3 Accuracy during discharge.

a. Conduct this test at the normal conditions specified in 4.3.1, except that the temperature tolerance shall be ± 2°F (1.1°C).
b. Discharge batteries at C/5 to cutoff voltage.
c. Charge batteries as specified in 4.6; use of 4.6.3 is not permitted.
d. Allow batteries to stabilize for not less than two hours.
e. Discharge batteries at C/5 to cutoff voltage and record capacity.
f. Charge batteries as specified in 4.6; use of 4.6.3 is not permitted.

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g. Allow batteries to stabilize for not less than two hours.
h. Discharge batteries at C/5 to cutoff voltage. Monitor and record SMBus state of charge information throughout the discharge where applicable.
    i. For display, non-smart, always on (see 3.5.8.1.1) visually monitor the state of charge indicators. Record the time of test when each state of charge indicator segment extinguishes (for user activated displays, it shall be activated not less than once every 20 minutes).
    ii. For display, non-smart, user activated (see 3.5.8.1.2) and display, smart (see 3.5.8.2) discharge the battery to approximately 90% state of charge. Activate the display. Record the time under test, SOC, and RTTE. Environmental chambers may be paused or shut off while accessing batteries to obtain the display readings. Repeat this step at approximately 70%, 50%, 30%, and 10% states of charge.
i. Repeat steps a through h at 14 ± 3.6ºF (−10 ± 2ºC) and 130 ± 3.6ºF (55 ± 2ºC) with a four hour soak prior to test.
j. Batteries shall meet the requirements of 3.5.8, 3.5.8.3, and 3.5.9.

4.7.2.14.4 Accuracy during charge. If the battery has a smart display (see 3.5.8.2), the following test shall be performed:

a. Conduct this test at the normal conditions specified in 4.3.1, except that the temperature tolerance shall be ± 2°F (1.1°C).
b. Discharge batteries at C/5 to cutoff voltage.
c. Charge batteries as specified in 4.6; use of 4.6.3 is not permitted.
d. Allow batteries to stabilize for not less than two hours.
e. Discharge batteries at C/5 to cutoff voltage.
f. Charge batteries as specified in 4.6; use of 4.6.3 is not permitted. Record the time to reach full state of charge.
g. Discharge batteries at C/5 to cutoff voltage.
h. While charging batteries as specified in 4.6 (use of 4.6.3 is not permitted), monitor and record SMBus state of charge information throughout the charge. Charge the battery to approximately 10% state of charge. Activate the display. Record the time under test and CTTF. Environmental chambers may be paused or shut off while accessing batteries to obtain the display readings. Repeat this step at approximately 30%, 50%, 70%, and 90% states of charge.
i. Repeat steps a through h at 14 ± 3.6ºF (−10 ± 2ºC) and 130 ± 3.6ºF (55 ± 2ºC) with a four hour soak prior to test.
j. Batteries shall meet the requirements of 3.5.8, 3.5.8.3, and 3.5.9.

4.7.2.15 Communication protocols.

4.7.2.15.1 SMBus. When specified (see 3.1), verify the following data has been programmed correctly using appropriate interface and software.

a. ManufacturerName
b. DeviceName (DoD type designation, e.g. BB-2590/U)

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c. DeviceChemistry
d. SerialNumber
e. ManufactureDate
f. DesignCapacity
g. DesignVoltage
h. ChargingCurrent
i. ChargingVoltage

4.7.2.15.2 USB power delivery. When specified (see 3.1), USB Power Delivery electrical compliance tests shall be performed in the appropriate PD2/PD3 and source/sink modes in accordance with USB Power Delivery Compliance Test Specification Version 1.0 Revision 1.2 (or later).

4.7.2.16 Concurrent charge/discharge. When specified (see 3.1), perform the charge and discharge tests. Batteries shall meet the requirements of 3.5.11.

4.7.2.17 USB power port connector(s). When specified (see 3.1), the following test shall be performed:

a. Verify that the USB Socket output is compliant with dedicated charging in accordance with the USB specification as specified.
b. Verify that the USB port can provide the specified current(s) and voltage(s) that this is provided concurrently with the specified load (see 3.1) on the battery connector. Batteries shall meet the requirements of 3.4.4.1.5 and 3.5.9.2.
c. Verify, wearing cold weather gloves, that the USB connector can be inserted with a mating USB plug without the presence of a plug in the battery connector.
d. Verify that the USB power port connector(s) are within the specified location and tolerances (see 3.1).
e. Verify that the USB connector(s) does not impede system operation (see 3.1).

4.7.2.18 Additional electrical requirements. When specified (see 3.1), batteries shall be subjected to additional specified electrical requirements.

4.7.3 Environmental tests. For each environmental test in 4.7.3.2 through 4.7.3.19, batteries shall be inspected for the defects listed in TABLE XIV prior to the start of the test.

4.7.3.1 Sequential testing. When verification of the requirements of 3.6 or capacity requirements of 3.5.3 are required for multiple environmental tests (see TABLE VII) performed in sequence on the same sample, single verifications may be used following the last test performed in that sequence. (Note: reducing inspection frequency may result in increased retesting in the event of a failure.)

4.7.3.2 Extreme low temperature discharge. For Type III (Li-ion) batteries the following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is not permitted. Store the batteries at -22 ± 2°F (-30 ± 1.1°C) for not less than 4 hours. Discharge under these

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conditions at the rate specified to the specified cutoff voltage (see 3.1). After testing, batteries shall meet the requirements of 3.5.3, 3.5.8, 3.6, and 3.6.1.

4.7.3.3 Extreme high temperature discharge. For Type III (Li-ion) batteries the following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is not permitted. Store the batteries at 131 ± 2°F (55 ± 1.1°C) for not less than 4 hours. Discharge under these conditions at the rate specified to the specified cutoff voltage (see 3.1). After testing, batteries shall meet the requirements of 3.5.3, 3.5.8, 3.6, and 3.6.1.

4.7.3.4 Altitude. The following test shall be performed. All battery types shall be subjected to the altitude test for lithium batteries of the UN Manual of Tests and Criteria. Prepare and fixture batteries as required by the manual prior to testing. Weigh batteries and record OCV prior to and after the specified conditions. Upon completion, visually examine the batteries in accordance with 4.7.1.1 and subject each battery in the fully charged state to the battery voltage tests of 4.7.2.2. Batteries shall meet the requirements of 3.6 and 3.6.2.

4.7.3.5 Temperature shock. The following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Subject batteries to temperature shock per Method 503, Procedure 1-B of MIL-STD-810 unless otherwise specified (see 3.1). The low temperature extreme (T1) shall be not greater than -75°F (-59°C) and the high temperature extreme (T2) shall be not less than 167°F (75°C). The stabilization periods shall be not less than 2 hours. Perform the capacity discharge test of 4.7.2.3. Batteries shall meet the requirements of 3.5.3, 3.5.8, 3.6, and 3.6.3.

4.7.3.5.1 Temperature shock high altitude high opening (HAHO) simulation. When specified (see 3.1), charge batteries in accordance with 4.6.1; use of 4.6.3 is permitted. Batteries HAHO test shall be performed in accordance with FIGURE 3.

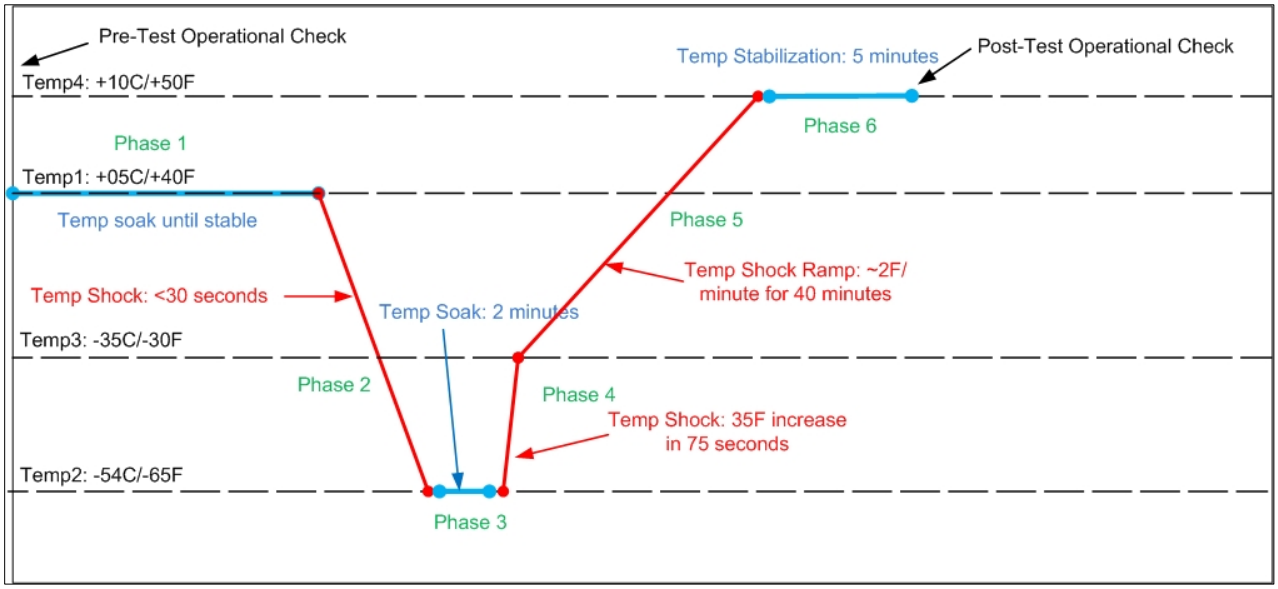


FIGURE 3. HAHO test profile

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4.7.3.5.2 Temperature shock high altitude low opening (HALO) simulation. When specified (see 3.1), charge batteries in accordance with 4.6.1; use of 4.6.3 is permitted. Batteries HALO test shall be performed in accordance with FIGURE 4.

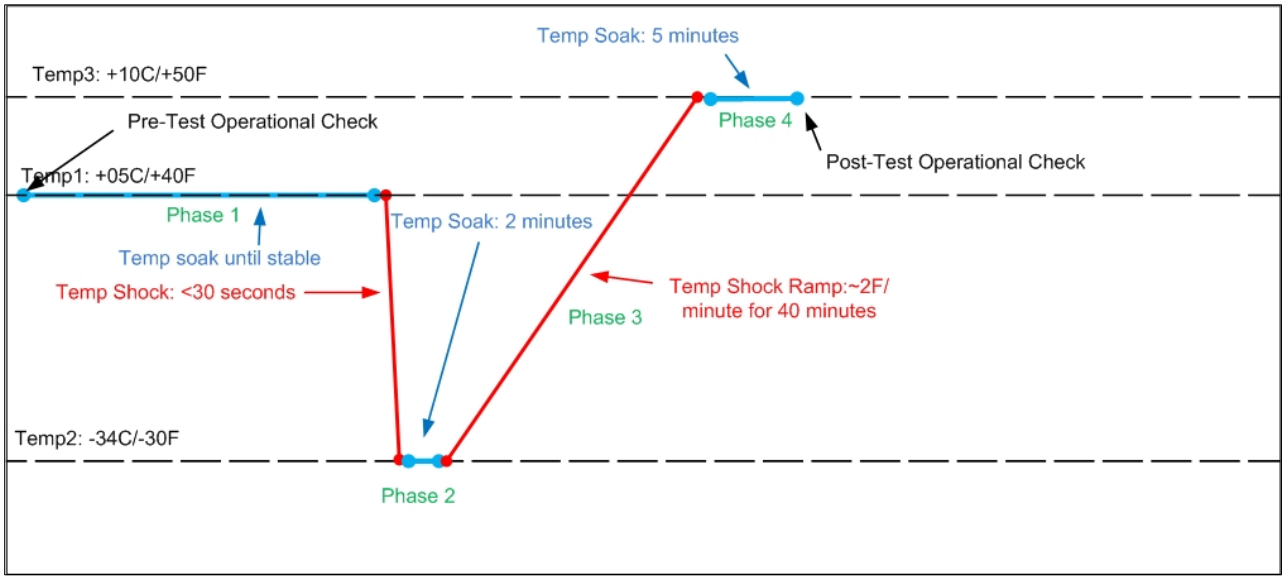


FIGURE 4. HALO test profile

4.7.3.6 Mechanical shock. The following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Rigidly attach the batteries to the test carriage by mounting screws or other appropriate hardware and shock test at not less than 40 gravity (g) with a pulse duration of not less than 18 milliseconds in accordance with test Method 516, Procedure 1, of MIL-STD-810. Apply three shocks per axis, on each of three orthogonal axes. Batteries shall meet the requirements of 3.5.3, 3.6, and 3.6.4.

4.7.3.7 Vibration. The following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. While discharging at the rate specified (see 3.1), batteries shall be subjected to minimum integrity vibration testing in accordance with MIL-STD-810 Method 514 using the general minimum integrity profile. Batteries shall meet the requirements of 3.6, and 3.6.5.

4.7.3.7.1 Vibration, loose cargo transportation. When specified (see 3.1), charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Batteries shall be subjected to loose cargo testing in accordance with MIL-STD-810 Method 514 Procedure II for not less than 20 minutes duration unless otherwise specified. Batteries shall be tested at ambient temperature unless otherwise specified. Batteries shall initially be placed stacked one on top of another in the battery tester with no other equipment present. After Step 4 of Procedure II, proceed to Step 9. Batteries shall meet the requirements of 3.6.5.

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4.7.3.8 Immersion.

4.7.3.8.1 Immersion, shallow. Weigh the batteries and record their weight. Test batteries in accordance with MIL-STD-810, Method 512, Procedure I. The test samples shall be fully charged and configured as specified:

a. Unless otherwise specified (see 3.1), duration shall be not less than 2 hours immersion at 3 feet each test.
b. For step 6 (Method 512, Procedure I), use the duration from 4.7.3.8.1a in saltwater per ASTM D-1141. For conformance inspection only (see 4.5), deionized water may be substituted for saltwater.
c. Battery temperature shall be 49°F (27°C) above the water temperature.
d. When specified (see 3.1), during the immersion period, verify that the batteries provide the specified load.
e. For step 9 (Method 512, Procedure I), conduct Capacity Discharge Test in 4.7.2.3. Batteries shall meet the requirements of 3.6.6.
f. After immersion, remove the batteries and dry exterior surfaces. Reweigh the batteries.
g. For Group A inspection only (see 4.5.1); duration shall be for not less than 30 minutes at one meter each test. For step 9 of MIL-STD-810, Method 512, Procedure I, perform a system check as specified. Batteries shall meet the requirements of 3.6.6.

4.7.3.8.2 Immersion, deep. When specified (see 3.1) batteries shall be tested as follows:

a. Weigh batteries and record their weight.
b. Immerse the batteries to an equivalent depth of not less than 33 feet in salt water substitute conforming to ASTM D-1141 for not less than 5 minutes.
c. Reweigh the batteries. Batteries shall meet the requirements of 3.6, 3.6.6, and the applicable specification sheet (see 3.1).
d. Repeat steps b and c with a depth of not less than 3 feet and time period of not less than 12 hours.

4.7.3.9 Transit drop.

4.7.3.9.1 Transit drop, normal. When specified (see 3.1), the following test shall be performed:

a. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
b. Stabilize batteries for 4 hours at 130 ± 2ºF (55 ± 1.1ºC).
c. Perform the drop at ambient conditions in not greater than 10 minutes after removal from the preconditioning temperature.
d. Drop each battery once from a height of not less than 30 inches onto cured concrete. The smallest side of the battery perpendicular to the plane of the connector face and nearest to the connector (where applicable) shall be parallel to the concrete surface and facing downward upon release, but need not be parallel upon impact. In the

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case of cylindrical batteries, the axis of the cylinder shall be parallel to the concrete surface upon release.

e. Visually examine batteries for the criteria in TABLE XIV.
f. Subject batteries to the battery voltage tests of 4.7.2.2 and the electrical leakage test of 4.7.4.7e.
g. Stabilize batteries for 4 hours at -20 ± 2ºF (-29 ± 1.1ºC).
h. Repeat steps c through f.
i. After test, batteries shall meet the requirements of 3.6 and 3.6.7

4.7.3.9.2 Transit drop, severe. When specified (see 3.1), the following test shall be performed. Batteries shall be dropped on each face, edge, and corner (see 3.1) (e.g. total of 26 drops for cubic batteries) for each temperature from a height not less than 48 inches onto steel backed by concrete specified in MIL-STD-810 Method 516 Procedure IV. When specified (see 3.1), batteries shall be dropped while mated to an object having the same mass, mass distribution, dimensions, and physical interface as the end item. Ten batteries shall be required for this test.

a. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted and verify that the State of Charge Indicators are properly functioning.
b. When specified (see 3.1), stabilize five of the batteries for 4 hours at 130 ± 2ºF (55 ± 1.1ºC) and the remaining five batteries for 4 hours at -20 ± 2ºF (-29 ± 1.1ºC).
c. Perform drops specified in TABLE XV as follows. When specified (see 3.1), perform drops 1 – 10 on unmated batteries. Perform the first requisite drop per battery at ambient conditions in not greater than 10 minutes after removal from the preconditioning temperature, and the final requisite drop not greater than 30 minutes after removal from the preconditioning temperature. If greater than 30 minutes are required, stabilize the battery for an additional 4 hours before proceeding.
d. When specified (see 3.1), subject batteries to the battery voltage tests of 4.7.2.2 and the electrical leakage test of 4.7.4.7e.
e. After test, batteries shall meet the requirements of 3.6 and 3.6.7.

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TABLE XV. *Transit drop, severe*

Drop	Impact 1/		Mated 2/	Sample number	
				-29 ºC	55ºC
1	Corner	top-back-left	YES	1	6
2	Edge	bottom-back	YES	1	6
3	Edge	top-right	YES	1	6
4	Edge	top-front	YES	1	6
5	Surface	right	YES	1	6
6	Corner	bottom-back-left	YES	2	7
7	Edge	front-right	YES	2	7
8	Edge	bottom-front	YES	2	7
9	Corner	top-front-left	YES	2	7
10	Corner	bottom-front-left	YES	2	7
11	Corner	top-back-right	NO	3	8
12	Edge	bottom-left	NO	3	8
13	Edge	top-back	NO	3	8
14	Surface	front	NO	3	8
15	Surface	left	NO	3	8
16	Corner	bottom-back-right	NO	4	9
17	Edge	bottom-right	NO	4	9
18	Edge	back-left	NO	4	9
19	Edge	front-left	NO	4	9
20	Surface	bottom	NO	4	9
21	Surface	back	NO	5	10
22	Surface	top	NO	5	10
23	Edge	top-left	NO	5	10
24	Corner	bottom-front-right	NO	5	10
25	Corner	top-front-right	NO	5	10
26	Edge	back-right	NO	5	10

NOTES

1/ To determine surface definitions, the battery shall be oriented with the main connector on the top. If the connector is off-center the battery shall be rotated so that the connector is in the back leftmost position.

2/ When specified (see 3.1)

4.7.3.10 Battery storage life. The following test shall be performed:

a. Discharge each battery at the 0.5C rate to the specified cutoff voltage. If the battery is Type III (Li-ion), proceed to the next step; if the battery is either Type I (Ni-Cd) or Type II (Ni-MH), skip step b and proceed to step c.
b. Charge each Type III battery with a constant current of not less than 100 milliamperes to not more than 220 milliampere-hours of capacity.
c. Store batteries at conditions of TABLE XVI ± 5°F for 28 cycles.
d. Return batteries to the normal conditions of 4.3.1 and store for not less than 2 hours.
e. Charge batteries at normal conditions in accordance with 4.6; use of 4.6.3 is permitted.
f. Perform the electric leakage test of 4.7.4.7d.

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g. Discharge each battery at the rate specified for capacity discharge of 4.7.2.3.
h. Repeat steps e, f, and g for two more cycles (a total of three charge/discharge cycles).
i. After test, batteries shall meet the requirements of 3.6 and 3.6.8.

TABLE XVI. *Battery storage life conditions*

Cumulative Test Time	Temp °C For reference only	Temp °F	Cumulative Test Time	Temp °C For reference only	Temp °F
1 Hr.	35	95	13 Hrs.	69	156
2 Hrs.	34	94	14 Hrs.	70	158
3 Hrs.	34	94	15 Hrs.	71	160
4 Hrs.	33	92	16 Hrs.	70	158
5 Hrs.	33	92	17 Hrs.	67	153
6 Hrs.	33	91	18 Hrs.	63	145
7 Hrs.	36	97	19 Hrs.	55	131
8 Hrs.	40	104	20 Hrs.	48	118
9 Hrs.	44	111	21 Hrs.	41	105
10 Hrs.	51	124	22 Hrs.	39	103
11 Hrs.	56	133	23 Hrs.	37	99
12 Hrs.	63	145	24 Hrs.	35	95

4.7.3.11 Impact. When specified (see 3.1), the following test shall be performed:
a. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted and verify that the State of Charge Indicators are properly functioning.
b. Configure the battery as specified.
c. Drop a 16 pound Professional Bowlers Association certified bowling ball or equivalent from not less than 36 inches onto the center-point of the battery ± one inch.
d. Repeat step c for the left and right sides of the horizontal centerlines of the battery not less than one half inch and not greater than two inches from the edge.
e. After test, batteries shall meet the requirements of 3.6 and 3.6.9.

4.7.3.12 Solar radiation. When specified (see 3.1) the following test shall be performed. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted and verify that the State of Charge Indicators are properly functioning. Test batteries per MIL-STD-810, Method 505, Procedure II using the spectral power distribution below (see TABLE XVII) for seven days. Position one battery with the front face up and one battery with the front face down. Wind speed shall be sufficient to maintain the battery surface temperature below 167°F (75°C). Below 149°F (65°C) surface temperature wind speed shall be 0 mph. Batteries shall meet the requirements of 3.6.10.

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TABLE XVII. *Spectral power distribution table*

Spectral Region	Bandwidth (nm)	Natural Radiation (% of total)	Tolerance (% of total)		Irradiance (W/m2)	Spectral Region Irradiance (W/m2)
			Min	Max		
Ultraviolet - B	280-320	0.5	0.3	0.7	5.6	5.6
Ultraviolet - A	320-360	2.4	1.8	3	26.9	62.7
	360-400	3.2	2.4	4.4	35.8	
Visible	400-520	17.9	16.1	19.7	200.5	580.2
	520-640	16.6	14.9	18.3	185.9	
	640-800	17.3	12.8	19	193.8	
Infrared	800-3000	42.1	33.7	50.5	471.5	471.5
				Totals:	1120	1120

4.7.3.13 Salt fog. When specified (see 3.1) the following test shall be performed. Test batteries per MIL-STD-810, Method 509 for four 24 hour cycles alternating 24-hour periods of salt fog exposure and drying conditions for not less than four 24-hour periods (two wet and two dry). The water used for the saltwater solution shall have a resistivity of not greater than 5 mega-ohms per cm (12.7 mega-ohms per inch). The battery shall meet the requirements of 3.6.11.

4.7.3.14 ESD. When specified (see 3.1), charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Subject the batteries to both the contact and air discharge test of IEC 61000-4-2, Level 4, configured as specified (see 3.1), connector shield, battery case, seams, and the digital display. The battery shall be tested with no load and a specified load (see 3.1). Verify that the specified load can be sustained for 1 minute following the test. Charge the battery as specified (see 3.1) and perform the discharge test of 4.7.2.3. The battery shall meet the requirements of 3.6.12.

4.7.3.15 Chemical resistance. When specified (see 3.1), batteries shall be sprayed separately with each of the following substances below. The substance shall cover not less than 50% of the surface area of the battery and include portions of all seams, state of charge indicator (if applicable), vent, label, charge terminal contacts, and connector. After standing for not less than 10 minutes, the batteries shall be rinsed with soap and water. Batteries shall meet the requirements of 3.6.13.

a. JP-8 per MIL-DTL-83133G or commercial diesel fuel.
b. Sodium Hypochlorite 5.25% (household bleach).

4.7.3.16 Rapid decompression. When specified (see 3.1) the following test shall be performed. Charge batteries in accordance with 4.6.2; use of 4.6.3 is permitted. Subject batteries to rapid decompression per Method 500, Procedure III of MIL-STD-810. Half of the batteries shall be tested with a mating connector and half shall have an exposed connector (i.e. without dust cap). Perform the capacity discharge test of 4.7.2.3. Batteries shall meet the requirements of 3.6.14.

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4.7.3.17 Explosive atmosphere. When specified (see 3.1), batteries shall be tested per the explosive atmosphere test of Method 511, Procedure I of MIL-STD-810. Batteries shall be configured as specified. Batteries shall meet the requirements of 3.6.15.

4.7.3.18 Combined temperature, altitude, and humidity. When specified (see 3.1), the following test shall be performed. Charge batteries in accordance with 4.6.2; use of 4.6.3 is permitted. Subject batteries to combined temperature, altitude, and humidity test for 3 cycles per TABLE XVIII.

No supplemental cooling shall be used. Performance check shall consist of open circuit voltage check according to 4.7.2.2. Post-test batteries shall meet the requirements of 3.5.3 and 3.6.

TABLE XVIII. *Combined temperature, altitude, and humidity*.

Test Phase Definition	Temperature °F (°C)	Relative Humidity	Altitude (ft)	Duration (min)
Cold Day Mission 1	≤ -26 (-32)	100%[1]	32000	60
Cold Day Mission 2	≤ -26 (-32)	100%[1]	32000	60
Cold Day Mission 3	≤ -26 (-32)	100%[1]	32000	60
**				
Transition to Hot				>20
Hot Day Mission 1	≥ 126 (52)	≤ 30%	32000	60
Hot Day Mission 2	≥ 126 (52)	≤ 30%	32000	60
Hot Day Mission 3	≥ 126 (52)	≤ 30%	32000	60
**				
Transition to Moist				>20
Warm Moist Day Mission 1	77±5 (25±2)	≥ 95%	32000	60
Warm Moist Day Mission 2	77±5 (25±2)	≥ 95%	32000	60
Warm Moist Day Mission 3	77±5 (25±2)	≥ 95%	32000	60
**				
Transition to Cold				>20

NOTES

[1]For chamber control purpose, 100% relative humidity implies as close to 100% relative humidity as possible, but not less than 95%.

4.7.3.19 Additional environmental tests. When specified (see 3.1), batteries shall be subjected to additional specified environmental tests.

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4.7.4 Safety tests.

4.7.4.1 Cell overcharge. The following test shall be performed. Weigh each cell to the nearest tenth of a milligram. Attach a thermocouple to the side of each cell to be tested, and attach current carrying and voltage monitoring leads to its terminals. Place each cell in a temperature chamber set at 77 ± 4.8°F (25 ± 2.7°C). Charge each cell at a constant current rate of C/2 for not less than 8 hours continuously. Measure and record cell temperature, voltage, and current. Visually examine the cells. Cells shall meet the requirements of 3.7.1.1 and 3.7.1.1.1.

4.7.4.2 Cell short circuit. The following test shall be performed. Weigh each cell to the nearest tenth of a milligram. Fully charge the cells in any appropriate manner at the normal conditions of 4.3.1. Stabilize cells at the same conditions after charge termination for not less than 1 hour. Short each cell individually by connecting the positive and negative terminals of the cell with a copper wire, not greater than 8 inches in length and with a cross-sectional area not less than American Wire Gauge (AWG) No. 16 wire. Discharge each cell to not greater than 0.2 volts. Visually examine the cells and weigh each cell to the nearest tenth of a milligram. Cells shall meet the requirements of 3.7.1.1 and 3.7.1.1.2.

4.7.4.3 Cell forced-discharge. The following test shall be performed. Weigh each cell to the nearest tenth of a milligram. Discharge each cell at the C/5 rate to not greater than 0.2 volts. Connect each cell in series with a 12-volt DC power supply and force discharge it at the C/5 rate for not less than 5 hours. Visually examine the cells and weigh each cell to the nearest tenth of a milligram. Cells shall meet the requirements of 3.7.1.1 and 3.7.1.1.3.

4.7.4.4 Nail penetration, cell. When specified (see 3.1), the following test shall be performed. The cells under test shall be recorded using video (at not less than 1080p 30 frames per second) and thermal cameras while subject to the following nail penetration tests. Recordings shall not cease until 10 minutes following the penetration of the last nail in the test.

Cells shall be charged to the specified state of charge (see 3.1). Insert a standard (see 6.12) 0.113 inch diameter stainless steel nail with a length that exceeds the width or diameter of the cell to be tested into each cell at right-angles to the electrode surface close to the center of the cell (± 15% of center of cell). Drive the nail completely through on one of the samples at a rate not less than 0.25 ft/sec; Drive the nail to a depth of 2/3 of the width of the cell for the other. Leave the nail in both cells for 24 hours. Record the date, time, place, ambient and cell temperatures (not less often than once per 15 seconds), and any events (e.g. smoke, fire, cell activity, and disassembly) along with the event time and associated event severity. Cells shall meet the requirements of 3.7.1.2 and shall be evaluated as part of the Safety Assessment Report in accordance with MIL-STD-882.

4.7.4.5 Crush, cell. When specified (see 3.1), the following test shall be performed. Subject cells designated for the crush test to the crush test of UL 1642. Crush each sample cell with its longitudinal axis parallel to the flat surfaces of the crushing apparatus. Subject each sample cell to a crushing force in only one direction. Cells shall meet the requirements of 3.7.1.3.

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4.7.4.6 Insulation resistance. The following test shall be performed. Apply a direct-current potential not less than 500 V between any two battery terminals not electrically connected and between battery terminals and the case of the battery. Measure the insulation resistance of batteries by the use of a copper plate making physical contact with the case. The plate shall exceed the dimensions of the surface to be tested. Place the battery under test on the plate so that the plate is visible outside all edges of the surface under test. Test each battery in not less than three locations. Test each battery on all surfaces with a case seam other than that on which the battery terminals are located; for cylindrical batteries, test in not less than three locations around the diameter including on seams if present. Batteries shall meet the requirements of 3.7.2.1.

4.7.4.7 Overcharge/electrical leakage. The following test shall be performed. After testing, batteries shall meet the overcharge and electrical leakage requirements (see 3.1).

a. Charge batteries in accordance with 4.6; use of 4.6.3 is not permitted.
b. Batteries shall be not less than 40 inches apart and the surface temperatures shall be monitored.
c. Overcharge the batteries at the rate specified (see 3.1) for not less than 21 hours. Measure and record battery voltages continuously or at a sampling rate of not less once per 60 seconds during overcharge.
d. Stabilize batteries at ambient conditions for not less than 2 hours.
e. Connect a 1 megaohm resistor to the following points: from the case or ground to each battery terminal and between all electrically isolated terminals (example: on a battery with two parallel sections, measure voltage from pin 1 to pins 2 and 5, from pin 4 to pins 2 and 5, and vice versa). Measure and record voltages at each connection point. Batteries shall meet the requirements of 3.7.2.2.
f. Discharge the batteries at the rate specified (see 3.1) to the cutoff voltage. Measure and record battery voltages continuously or at a sampling rate of not less than once per 60 seconds during discharge.
g. Repeat steps a through f for a total of 10 cycles.
h. Visually examine the batteries for the criteria of TABLE XIV.
i. Examine batteries for the requirements of 3.6.
j. Examine batteries to insure compliance with the requirements of 3.5.3.
k. If additional charge contact are present, repeat steps a through j for one cycle for each set of charge contacts.

4.7.4.8 Short circuit protection. The following test shall be performed. Charge batteries as specified in 4.6; use of 4.6.3 is permitted. Measure and record the OCV. Short each battery across all the positive and negative terminals with a total external resistance not greater than 50 milliohms. After one hour remove the short from across the terminals. Measure and record the OCV. Stabilize batteries at the normal conditions of 4.3.1 for not less than 2 hours. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted. Stabilize batteries at normal conditions for not less than 2 hours, then discharge the battery in accordance with 4.7.2.3. The battery shall meet the requirements of 3.7.2.3.

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4.7.4.9 High temperature temporary cutoff. The following test shall be performed. Charge batteries as specified in 4.6; use of 4.6.3 is permitted.

a. Store the batteries at 167 ± 7°F (75 ± 5°C) for not less than 2 hours. Measure and record the battery voltage.
b. Attempt to charge the battery in accordance with 4.6; use of 4.6.3 is permitted. Measure and record any resulting current.
c. If the battery has a smart display (see 3.5.8.2), verify that the battery shows that it is in an unusable condition.
d. Attempt to discharge the battery in accordance with 4.7.2.3. Measure and record any resultant current from either attempt. Measure and record any resulting current.
e. Allow the battery to cool down to 104 +0/-9°F (40 +0/-5°C). Measure and record battery voltage.
f. Repeat steps a through e five (5) times.
g. Perform the capacity discharge test of 4.7.2.3.
h. Batteries shall meet the requirements of 3.7.2.4.

4.7.4.10 High temperature permanent cutoff. The following test shall be performed. Charge batteries as specified in 4.6; use of 4.6.3 is permitted.

a. Store the batteries at 203 ± 9°F (95 ± 5°C) for not less than 2 hours. Measure and record the battery voltage.
b. Attempt to charge the battery in accordance with 4.6; use of 4.6.3 is permitted. Measure and record any resulting current.
c. Attempt to discharge the battery in accordance with 4.7.2.3. Measure and record any resultant current from either attempt. Measure and record any resulting current.
d. Allow the battery to cool down to 104 +0/-9°F (40 +0/-5°C). Measure and record battery voltage.
e. Repeat steps b and c.
f. Batteries shall meet the requirements of 3.7.2.5.

4.7.4.11 Projectile. When specified (see 3.1), the following test shall be performed. The batteries under test shall be recorded using video (at not less than 1080p 30 frames per second) and thermal cameras while subject to the following projectile test. During test, batteries must be secured to the screen with a single wire tied around the sample. Recordings shall not cease until 10 minutes following the test. Subject each battery designated for the projectile test to the projectile test of UL 1642. Record the date, time, place, ambient and battery temperatures (not less often than once per 15 seconds), and any events (e.g. smoke, fire, cell activity, and disassembly) along with the event time and associated event severity. Batteries shall meet the requirements of 3.7.2.8 and shall be evaluated as part of the Safety Assessment Report in accordance with MIL-STD-882.

4.7.4.12 Nail penetration, battery. When specified (see 3.1), the following test shall be performed. Insert a not narrower than a standard (see 6.12) 0.113 inch diameter stainless steel nail with a length that exceeds the width or diameter of the battery. The rate of the nail shall be not less than 0.25 ft/sec and penetrate though the width of the battery. Leave the nail in the

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battery for 24 hours. Record the date, time, place, ambient and battery temperatures (not less often than once per 15 seconds), and any events (e.g. smoke, fire, battery activity, and disassembly) along with the event time and associated event severity. The batteries under test shall be recorded using video (at not less than 1080p 30 frames per second) and thermal cameras while subject to the following nail penetration tests. Recordings shall not cease until 10 minutes following the penetration of the last nail in the test. Batteries shall meet the requirements of 3.7.2.9 and shall be evaluated as part of the Safety Assessment Report in accordance with MIL-STD-882.

4.7.4.12.1 Single penetration test.

a. Charge the battery in accordance with 4.6; use of 4.6.3 is permitted.
b. Place the batteries on a surface that will not inhibit the movement of the nail
c. Mark a random location (see 6.11) on each test sample to be penetrated.
d. Batteries shall be placed on a specified load (see 3.1).
e. Use a specified stainless steel nail with a length that exceeds the width or diameter of the battery. Drive the nail through the samples so that the nail protrudes through both sides of the battery at a rate not less than 0.25 ft/sec.
f. Record for each test: the date, time, place, ambient and battery temperatures (not less often than once per 15 seconds), and any events (e.g. smoke, fire, cell activity, and disassembly) along with the event time and associated event severity.
g. Batteries shall meet the requirements of section 3.7.2.9 and shall be evaluated as part of the Safety Assessment Report in accordance with MIL-STD-882.

4.7.4.12.2 Multiple nail penetration test.

a. Charge the battery in accordance with 4.6; use of 4.6.3 is permitted.
b. Place the batteries on a surface that will not inhibit the movement of the nail.
c. Randomly mark (see 6.11) no less than three locations on each test sample to be penetrated.
d. Batteries shall be placed on a specified load (see 3.1).
e. Use a specified stainless steel nail with a length that exceeds the width or diameter of the battery. Drive the nail completely through the samples.
f. Record for each test: the date, time, place, ambient and battery temperatures (not less often than once per 15 seconds), and any events (e.g. smoke, fire, cell activity, and disassembly) along with the event time and associated event severity.
g. The multiple penetrations shall occur in not greater than 1 second from first to last penetration.
h. Batteries shall meet the requirements of section 3.7.2.9 and shall be evaluated as part of the Safety Assessment Report in accordance with MIL-STD-882.

4.7.4.13 Weapons characterization. The following test shall be performed (see 6.13) for weapon / bullet impact characterization. The batteries under test shall be recorded using video (at not less than 1080p 30 frames per second) and thermal cameras while subject to the following tests. Recordings shall not cease until 10 minutes following the last test. For each test record the date, time, place, ambient and battery temperatures, (not less often than once per 15

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seconds), and any events (e.g. smoke, fire, cell activity, and disassembly) along with the event time and associated event severity.

a. Charge batteries in accordance with 4.6; use of 4.6.3 is permitted.
b. Orient the battery as specified (see 3.1).
c. Mark battery locations most likely to elicit a reactive response (see 6.13.1). Ensure, at a minimum, one cell will be impacted.
d. Connect the battery to the specified load (see 3.1). Not less than 2 minutes prior to impact, record power output with a frequency of not less than once every 30 seconds.
e. While continuing to record the output per step d, subject batteries to specified threats (see 3.1) targeting the spots marked in step c.
f. Continue recording per step d for not less than 2 minutes following the last impact.
g. Record battery output with a specified load for not less than 4 minutes after each penetration to categorize power output capability post impact.
h. At the completion of each test, verify compliance with 3.7.2.11.
i. Power output (if any) shall be recorded for not less than eight minutes at a frequency of not less than once every 30 seconds.

4.7.4.14 Additional safety tests. When specified (see 3.1), batteries shall be subjected to the additional specified safety tests.

4.7.5 Lithium battery safety program (US Navy). When specified (see 3.1) the following tests shall be performed on Types III batteries. Test each battery in a test vessel (see 6.14.2) in order to facilitate pressure readings in lieu of the “complete system” described by NAVSEA S9310-AQ-SAF-010. Battery voltage, charge or discharge current, and test vessel pressure shall be monitored continuously during testing. Test samples will be subjected to testing in accordance with TABLE XIX. Batteries shall be evaluated as part of the Safety Assessment Report in accordance with MIL-STD-882.

TABLE XIX. *Lithium battery safety program sample size*

Inspection	Sample size	Requirement Paragraph	Test Method Paragraph
Visual mechanical	12	3.4	4.7.1.1
Battery open circuit voltage	12	3.5.2	4.7.2.2
Short circuit test	3	3.7.2.10	4.7.5.1
Overcharge/discharge test	3	3.7.2.10	4.7.5.2
Overcharge/charge test	3	3.7.2.10	4.7.5.3
Electrical safety device test	3	3.7.2.10	4.7.5.4

4.7.5.1 Short circuit. When specified (see 3.1), the short circuit test for rechargeable batteries of NAVSEA S9310-AQ-SAF-010 shall be performed. Batteries shall meet the requirements of 3.7.2.10.1.

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4.7.5.2 Overcharge/discharge. When specified (see 3.1), the overcharge/discharge test for rechargeable batteries of NAVSEA S9310-AQ-SAF-010 shall be performed. Batteries shall meet the requirements of 3.7.2.10.1.

4.7.5.3 Overcharge/charge. When specified (see 3.1), the overcharge/charge test for rechargeable batteries of NAVSEA S9310-AQ-SAF-010 shall be performed. Batteries shall meet the requirements of 3.7.2.10.1.

4.7.5.4 Electrical safety device. When specified (see 3.1), the electrical safety device test for rechargeable batteries of NAVSEA S9310-AQ-SAF-010 shall be performed. Batteries shall meet the requirements of 3.7.2.10.2.

4.7.6 Interoperability.

4.7.6.1 Charger interoperability. A battery charger interoperability test shall be performed. Batteries shall be charged on currently fielded standard Army chargers and adapters. Verify no anomalies occur during charging and batteries meet 90% of their rated capacity and meet the requirements of 3.10.

4.7.6.2 End item interoperability. When specified (see 3.1), batteries shall be subjected to specified end item interoperability.

4.7.7 Electromagnetic interference. When specified (see 3.1), batteries shall be subjected to specified electromagnetic verifications.

4.7.8 Additional test methods. When specified (see 3.1), batteries shall be subjected to the additional specified test methods.

5 PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Sealed, high energy density, rechargeable batteries covered within this specification are used in military electronics and communications equipment, electric motors, and for other military power requirements. These batteries are subjected to severe environmental conditions.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of the specification.
b. The specific issue of individual documents referenced, if required (see 2).
c. Packaging requirements (see 5.1).
d. When first article inspection and rough handling tests of equipment are required. If certification of parts, materials, or components are required (see 3.2.1).
e. Actions required relative to inspection failure.
f. Any additional case markings, dependent upon battery chemistry.
g. Tests, if any, requiring government personnel or facilities.
h. Delivery schedule (impacts inspection lot, see 6.8).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.4 Definitions.

6.4.1 Complete cell assembly (CCA). A CCA is the total number of cells used within a complete battery along with the intercell connections; for testing purposes, it needs to be connected to the ECA as a separate item in order to facilitate sensing connections between the two assemblies.

6.4.2 Cutoff voltage. The cutoff voltage of the battery is the specified final voltage reached at the end of discharge.

6.4.3 Cycle. A cycle is defined as a battery being subject to four phases: 1) Charge 2) Rest 3) Discharge, and 4) Rest.

6.4.4 Discharged batteries. Batteries will be considered discharged when a specified discharge rate was conducted until a specified cutoff voltage was reached.

6.4.4.1 Discharge rate. The rate at which energy is removed from a charged battery.

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6.4.4.2 C (rate).

$$I = C/t$$

where

I = current in amperes
C = rated ampere hour capacity
t = hourly rate of discharge

For example, the expression for the five hour discharge current for a 25 ampere hour battery will be I = (1/5)(25) = 5 amperes.

6.4.5 Electronic component assembly (ECA). An ECA is the total of all electronic components, connectors and mounting boards used within a complete battery.

6.4.6 Fully charged batteries. Batteries will be considered fully charged when charged in accordance with 4.6. In addition, the SOCI must indicate a fully charged battery.

6.4.7 Level 2 and 3 chargers. Smart Battery Charger Types are described by the Smart Battery Charger Specification as:

Level 2 – Smart Battery Controlled Smart Battery Charger
Level 3 – Host Controlled Smart Battery Charger

Refer to the Smart Battery Charger Specification for a more complete description. Copies of this document may be obtained from the System Management Interface Forum (SMIF), Inc, on-line at http://powersig.org.

6.4.8 Lithium-ion. Lithium-ion batteries are comprised of cells that contain lithium compounds and exchange lithium ions between electrodes. This does not include batteries containing pure lithium metal in the anode.

6.4.9 Non-flammable and non-toxic materials. Non-flammable and non-toxic materials are those materials which will not support combustion, produce smoke, or be capable of emitting toxic fumes, in a cured state, when subjected to the environmental conditions specified herein.

6.4.10 Sleep mode. State of reduced power consumption of electronic circuity within the battery.

6.4.11 Softening temperature. The softening temperature of insulation is the temperature at which it will distort under a reasonable amount of pressure when tested per the Heat Distortion Test of ASTM D 2633.

6.5 Technology insertion. While this specification sheet is written to be technologically neutral, requirements for individual batteries (e.g. safety, voltage, current, etc.) must be compatible with existing infrastructure to include end items and chargers. This may limit

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available technologies for existing batteries. For example, if existing cells have a voltage of 4.2 V, a cell with a voltage of 5 V (due to an emerging technology) may not be compatible.

6.6 Approved chargers. The following are the approved chargers that may be used for the charge method of 4.6.3 above. Note that some of these chargers may be of an older vintage and may not charge every battery described by the associated specification sheets. Users of this document will need to assure compatibility with the products under test prior to charging in accordance with 4.6.3. Individual specification sheets may list other chargers or not allow chargers appearing on this list. This listing is provided for reference only and does not constitute an endorsement of the products by the DoD.

TABLE XX. *Approved chargers*

TYPE DESIGNATION	COMMON NAME	NIIN	PART NUMBER	CAGEC
PP-8498/U	Soldier Portable Charger (SPC)	01-495-2839	BTC-70801	51828
PP-8481B/U	Vehicle Mounted Charger (VMC)	01-527-2726	BTC-70836	51828
N/A	Advanced Battery Charger (ABC)	01-660-3696	BTC-70100	51828
N/A	Universal Battery Charger (UBC)	01-659-7090	4102871-501	23386
N/A	Standard UBC-Lite with CWB Charging	01-687-8917	1600903-1	23386
N/A	Standard UBC-Lite Complete kit	01-687-3120	1100771-503	23386
N/A	Squad Power Mgr.	01-626-9656	SPM-622	3B0F7

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.8 Inspection lot. The inspection lot is the quantity of batteries (exclusive of the number of batteries required as samples) produced at any one place of manufacture on any one contract delivery order presented to the Government for acceptance. In larger battery buy orders, it would be expected that the Contractor would have multiple inspection lots to fulfill the buy order each of which would complete conformance acceptance testing as required in the statement of work associated with said contract.

6.9 Test report data. The following data points will be necessary for Government review and should be included in First Article and Conformance Test Reports:

a. Certifications, where required (see 3.2.1).
    i. Supporting documentation used for certification (e.g. Component data sheets)

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b. Whenever there is a service or ampere hour requirement, include the following:
    ii. Initial Open Circuit Voltage (IOCV)
    iii. Initial Loaded Voltage (ILV)
    iv. Battery or Cell Voltage and Current, recorded at regular intervals. Interval should be Service Requirement or shorter.
    v. The time elapsed between initial voltage and cutoff voltage.
    vi. Record how low a battery is discharged in volts, particularly whenever near zero volts.
    vii. Record date and time of day that any portion of test is initiated in the test report including initiation of soak at test temperature.
    viii. When a requirement limit is specified, record the final quantitative figure.
c. Photographs and videos as applicable.

6.10 First Article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 4.4), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids or requests for proposal should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.11 Random variables. The DoD may require a review or assignment of any random variables. Potential offerors are cautioned to review solicitations for such requirements.

6.12 Standard nail. The construction industry uses nail guns with nails of standard diameter sizes. The reliance on manufacturing standards for diameter is sufficient.

6.13 Weapons characterization.

6.13.1 Bullet location. The DoD may require the battery to be evaluated by Government officials and bullet locations to be assigned accordingly. Potential offerors are cautioned to review solicitations for such requirements.

6.13.2 Government participation. The DoD may require the weapons characterization test to be conducted at a Government facility and/or have a designated Government official be present during the test. Potential offerors are cautioned to review solicitations for such requirements.

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6.14 Lithium Battery Safety Program (US Navy). Potential offerors and contractors should be aware that passing the test criteria specified herein for the US Navy Lithium Battery Safety Program will not constitute a “Safety Approval” for the battery by the US Navy. In accordance with S9310-AQ-SAF-010, Navy department users must still apply for battery safety approvals based on the NSN of the battery intended for use, as well as both the intended Navy platform and each specific using end item.

6.14.1 Test data. The intent of including these test requirements in this specification are primarily to give manufacturers visibility into the essential safety characteristics needed by the Navy. Until now, the tests were performed in evaluating system or device safety without any knowledge of these requirements by battery manufacturers. Additionally, these requirements will provide the Navy with test data on file when assessing applications for safety approvals. The data provided will be used for engineering assessments that are capable of providing recommendations for safer battery compartments in battery-using devices.

6.14.2 Test vessel. Due to the need to record pressure changes during test, the Navy safety tests require use of a test “vessel”, defined as a sealed container larger than the battery under test. A “sealed container” could be something as simple as an unused paint can with the lid on, provided that gaskets, plugs, or sealants are used around the access points for load and sensing leads to preserve the seal. A calculation of the remaining void space inside the vessel containing the battery under test will be needed for proper interpretation of the pressure readings.

6.14.3 Navy test facility. The DoD has a preference for having the tests described by S9310-AQ-SAF-010 (see 4.7.5) conducted in a US Navy test facility. Potential offerors are cautioned to review solicitations for such requirements. Should testing at a Navy facility be required, offerors will need to obtain a quote for Navy testing from the Navy in order to respond to the solicitation. Points of Contact in the Navy for the quote will be identified in the solicitation documentation. Offerors will need to assure that Navy testing costs are covered in their price proposal. Successful offerors will need to establish a Test Service Agreement (TSA) with the Navy after award. Unsuccessful offerors will be under no obligation to pay for anything associated with the quote obtained earlier. TSAs have expiration dates; if an agreement is already established, check to make sure that it will still be in effect at the time when testing may be required.

6.15 Interoperability.

6.15.1 Charger interoperability. The DoD has a preference for having the tests described by 4.7.6.1 conducted in a US Army battery test facility. This test is to ensure backwards compatibility with existing fielded chargers. Batteries may be tested in various degraded states such as an overheated state, various contacts occluded, etc. Potential offerors are cautioned to review solicitations for such requirements.

6.15.2 End item interoperability.

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The DoD has a preference for having the tests described by 4.7.6.2 conducted in a US Army test facility. This test is to ensure backwards compatibility with existing end items. Potential offerors are cautioned to review solicitations for such requirements.

6.16 Electromagnetic interference. Some of the batteries described by this specification are required to be electromagnetically compatible with specific systems (see the applicable specification sheet for system identity). This compatibility is specific to the system identified and should not be assumed for other systems as well. Any other system using such a battery will need to have its electromagnetic compatibility with the battery evaluated separately, and in the event the battery is not compatible, then either system modifications will be needed or the system will need another battery.

6.16.1 Government participation. The DoD may require the electromagnetic compatibility test to be coordinated with the appropriate Government office. Potential offerors are cautioned to review solicitations for such requirements.

6.17 Shelf life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85% shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, Shelf-life Management Manual. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.18 Nominal ratings. See 3.1.

6.19 Subject term (key word) listing.

Advanced chemistry
Bullet
Energy
Flammable
Lithium ion
Nickel cadmium
Nickel metal hydride
Polymer
Power
Shelf-life
Weapons characterization

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6.20 Additional notes. See 3.1.

Custodians:
Army – CR
Navy – MC

Preparing Activity
Army – CR
(Project Number 6140-2020-005)

Review activities:
Navy – AS,SH
Air Force – 71
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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